SHETH L.U.J & SIR MV COLLEGE

Aim:Analyzing Data using Table Calculations in Tableau

a) Apply quick table calculations (running total, percent of total)
b) Create moving averages and cumulative metrics
c) Perform year-over-year growth analysis
d) Use ranking functions (Top N, Bottom N)
e) Implement window functions (WINDOW_SUM, WINDOW_AVG)
f) Customize addressing and partitioning
g) Create dynamic titles and labels using table calculations

A: Apply quick table calculations (running total, percent of total)

Quick Table Calculations allow you to apply complex computation logic (like cumulative sums

or proportions) to your visualizations instantly without manually writing code.

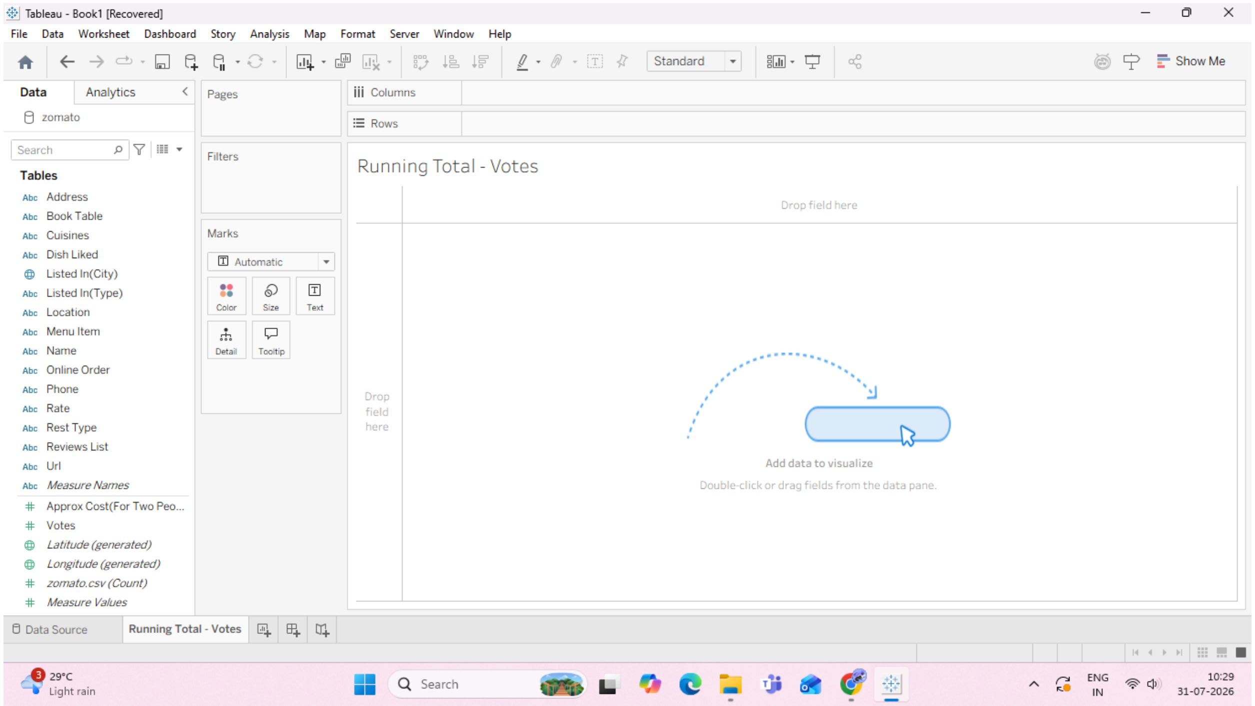
Task 1: Create a Running Total of Votes by City

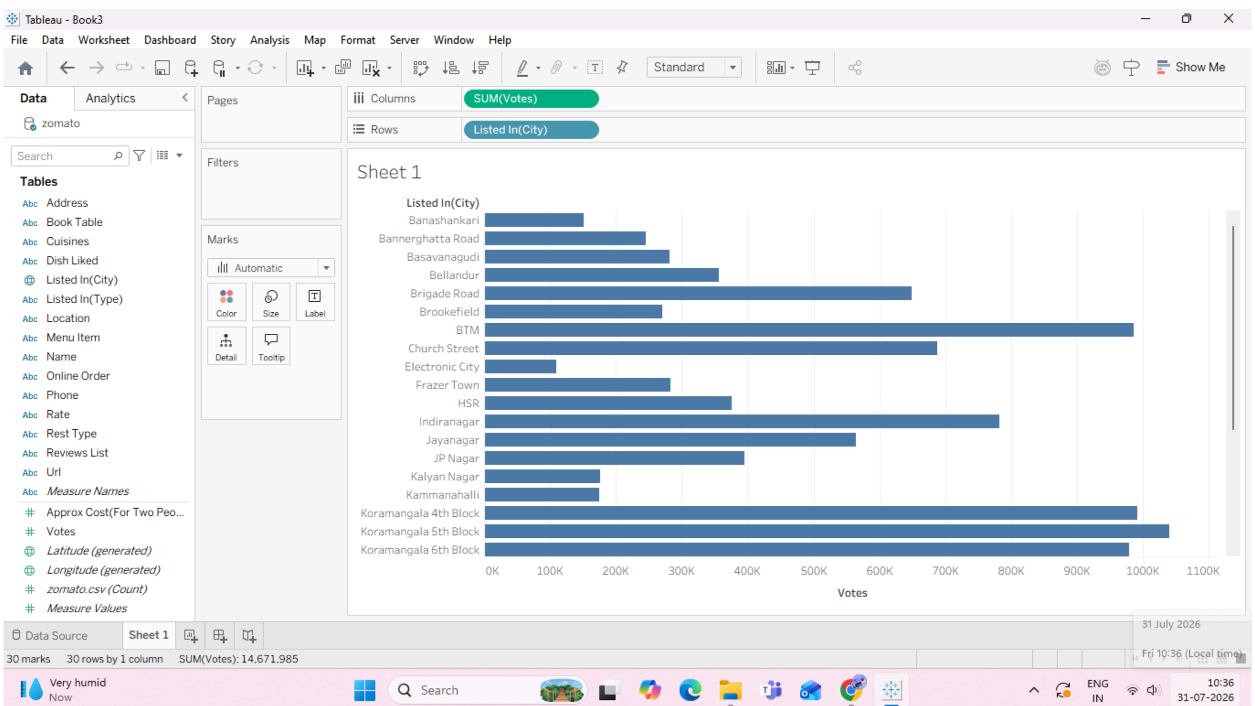
A Running Total calculates a cumulative sum across a dimension (e.g., adding each city’s votes

to the running sum of all previous cities).

SHETH L.U.J & SIR MV COLLEGE

1. Open a new worksheet and rename it: Running Total - Votes.

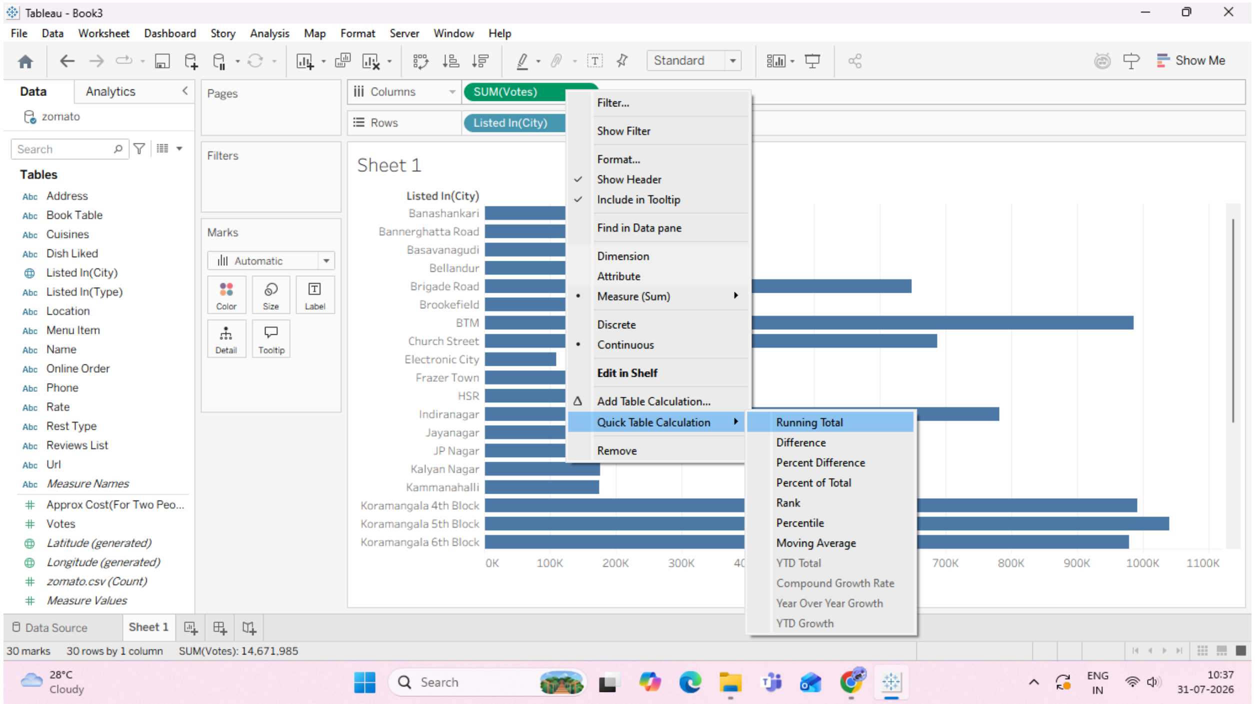




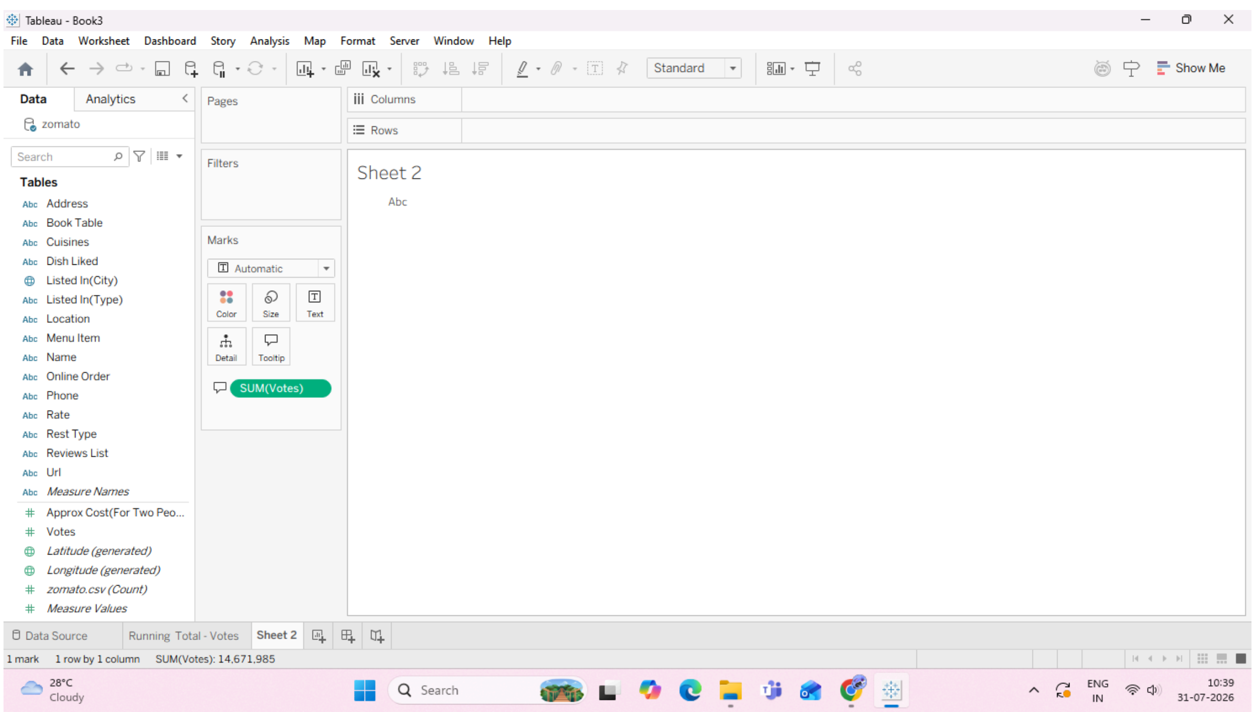
4. Convert to Quick Table Calculation:

Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

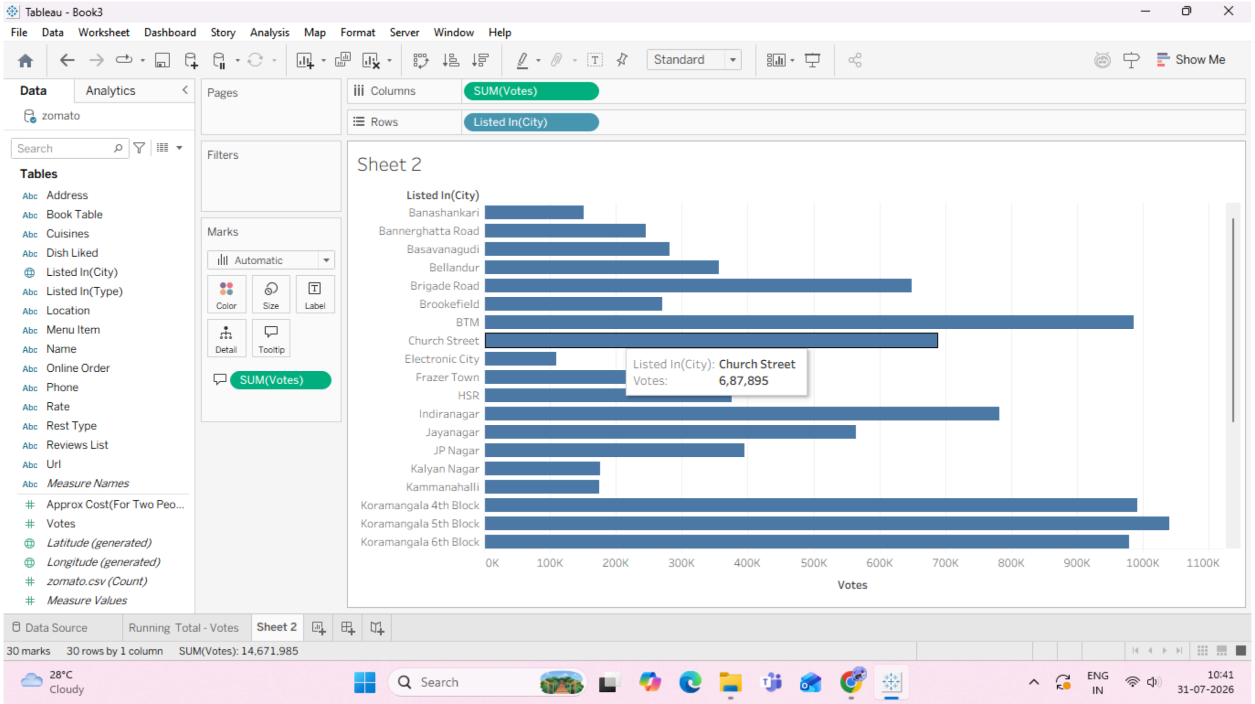


5. Add Visual Context (Secondary Measure):

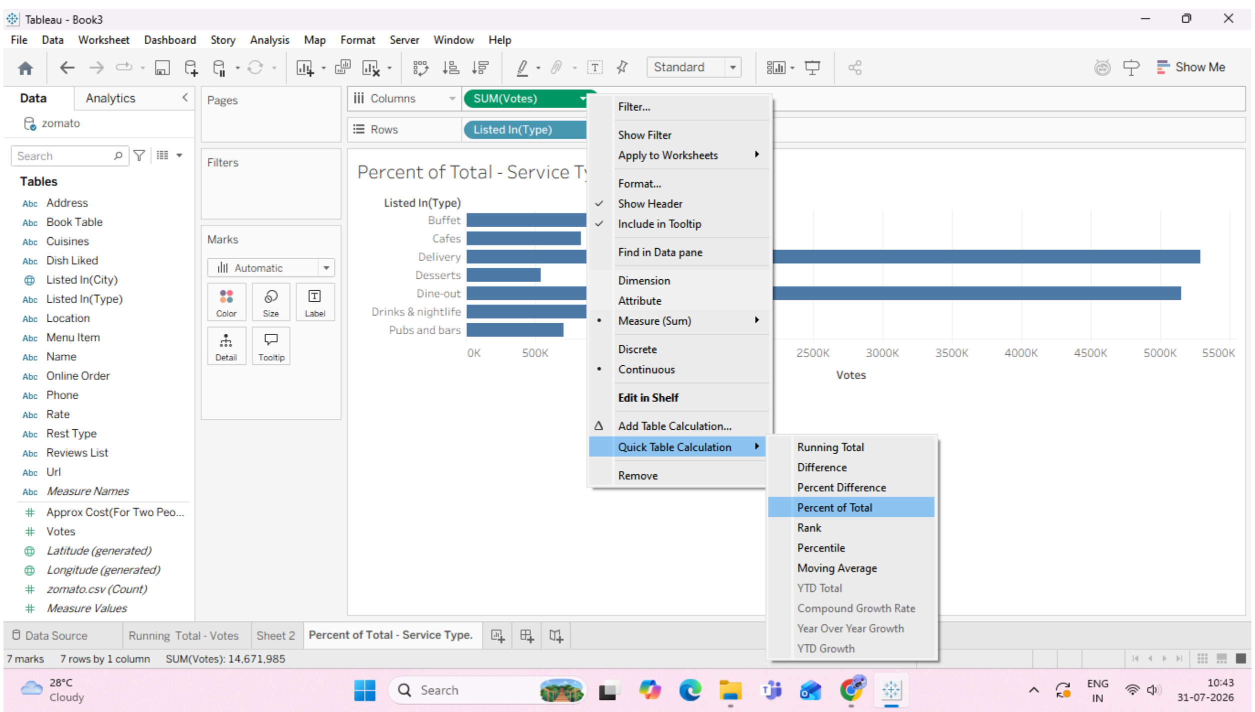


Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

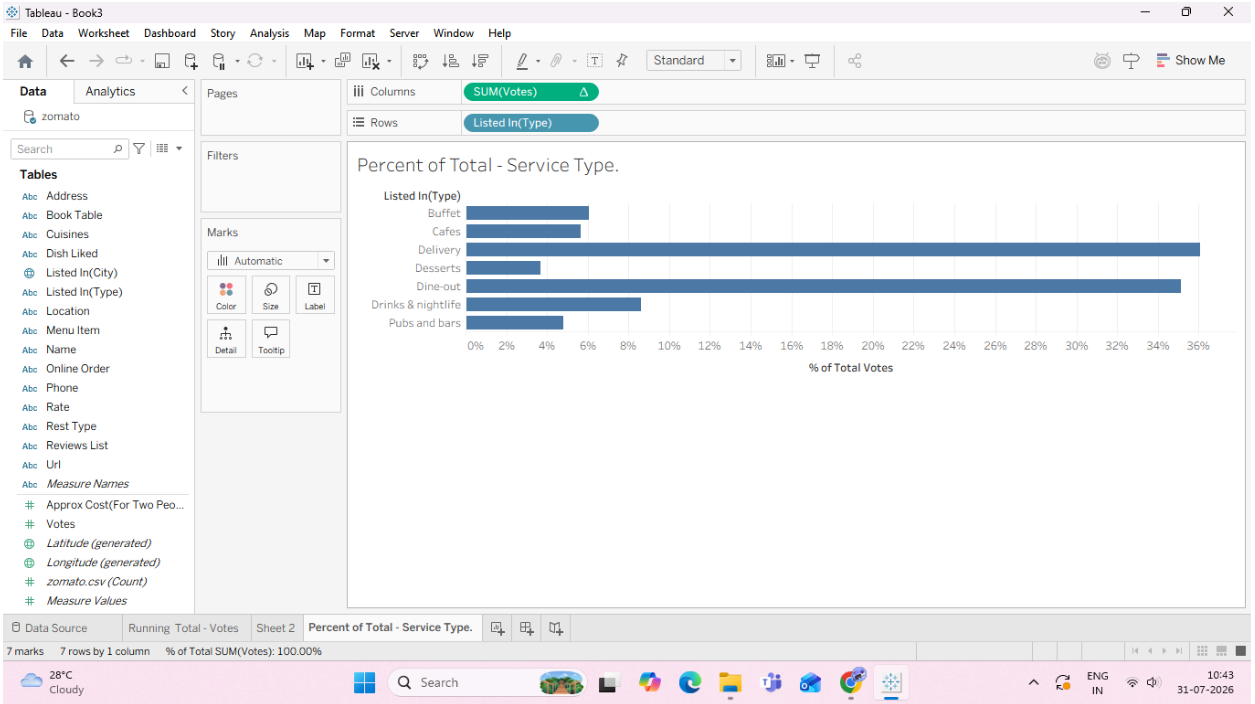


Task 2: Create a Percent of Total across Categories A Percent of Total transforms raw numbers into relative percentages of the entire view (summing to 100%).

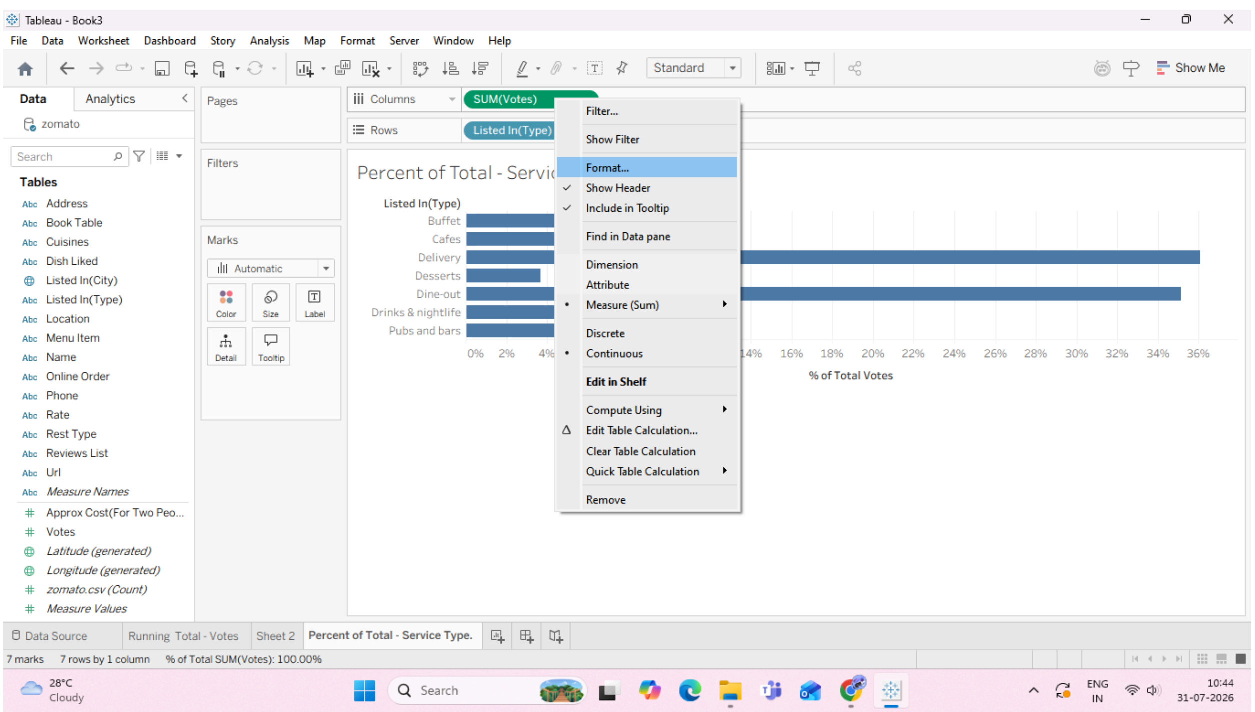


Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

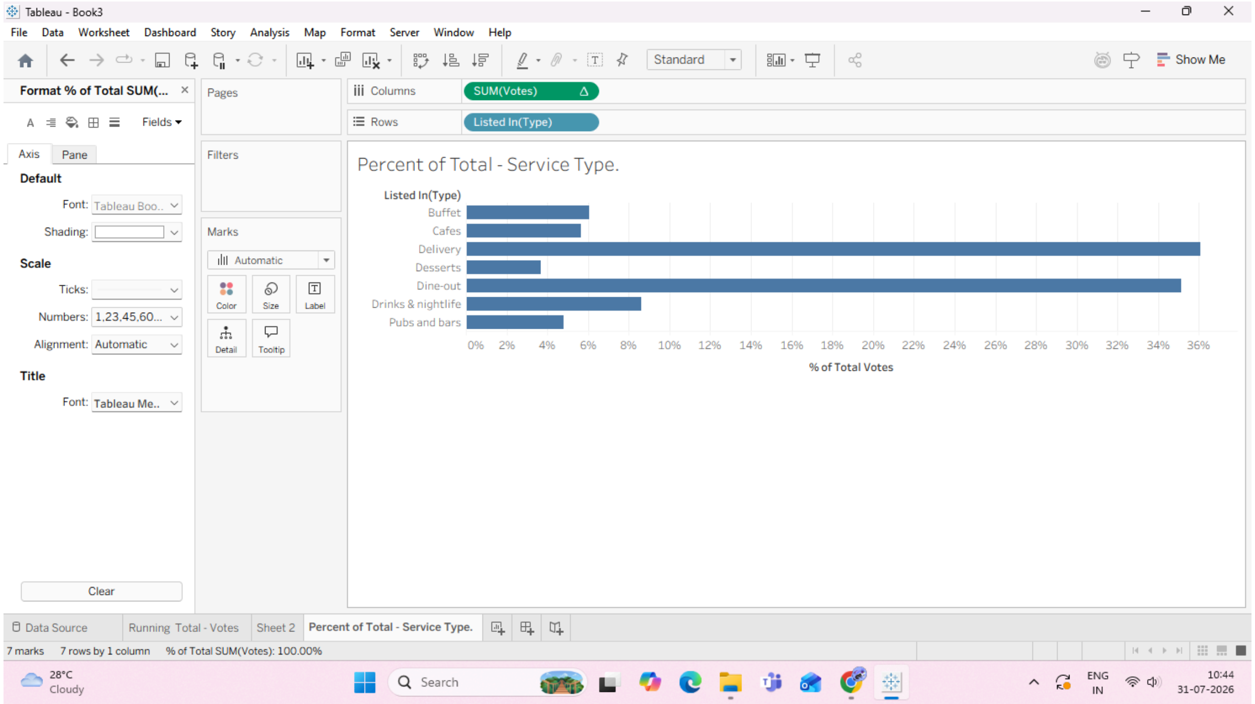


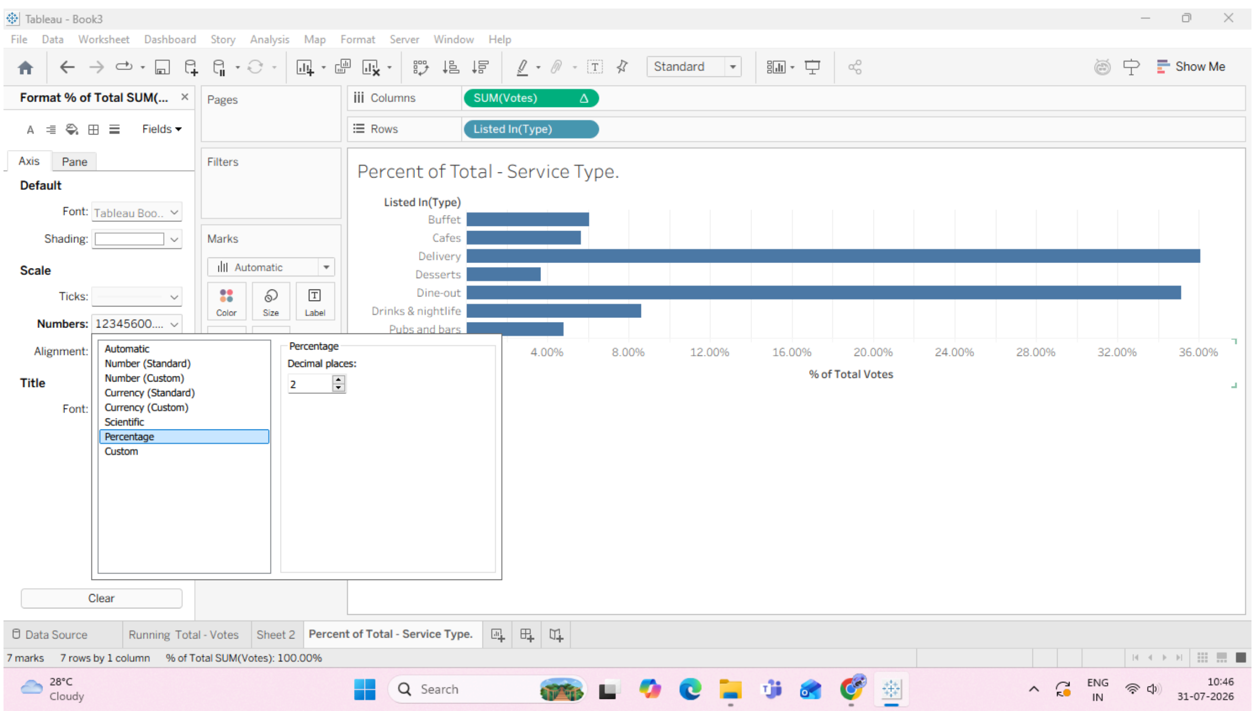
6. Format as Percentage:



Purvi.R.Karkera
T086

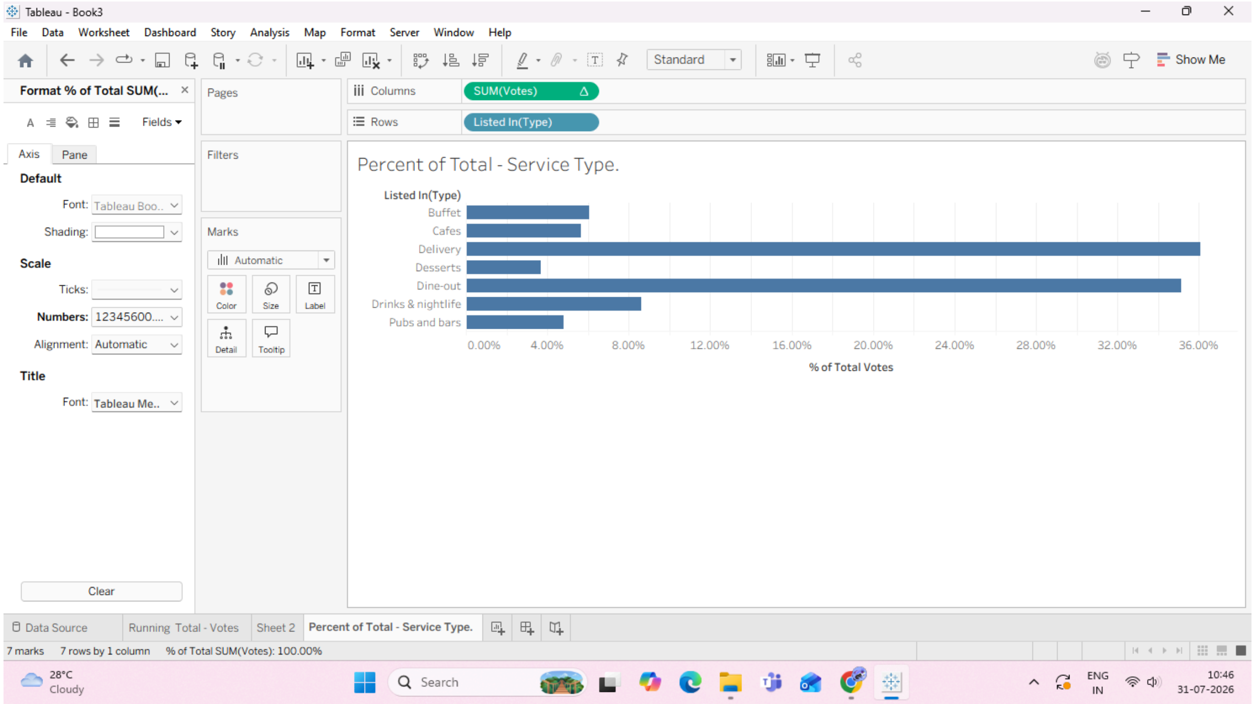
SHETH L.U.J & SIR MV COLLEGE



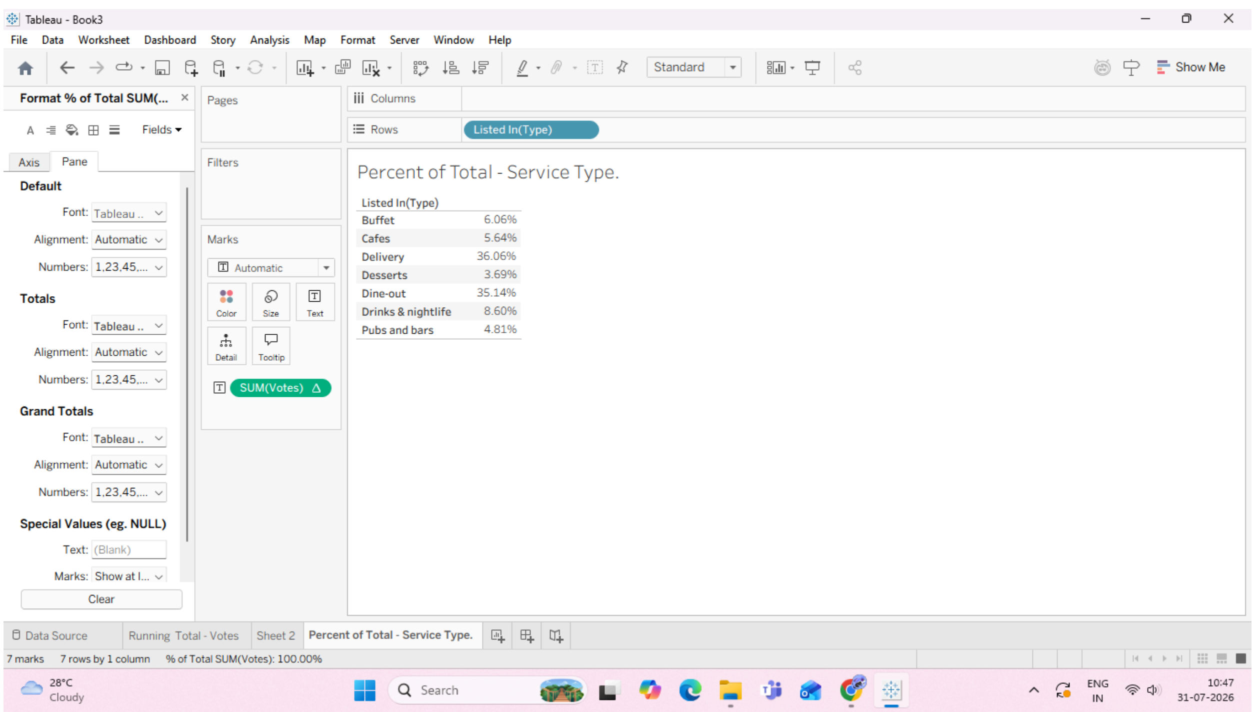


Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE



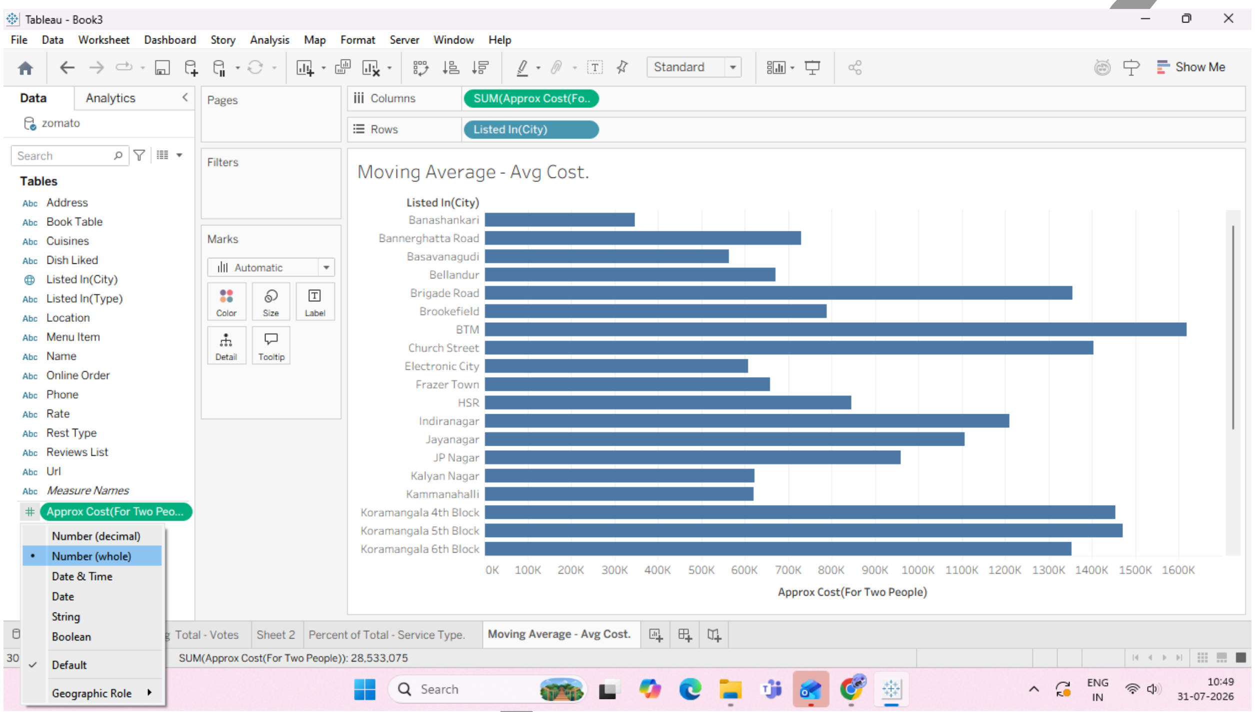
7. Drag SUM(votes) (with the table calculation icon Δ) onto the Label mark so the exact percentages display on top of each bar



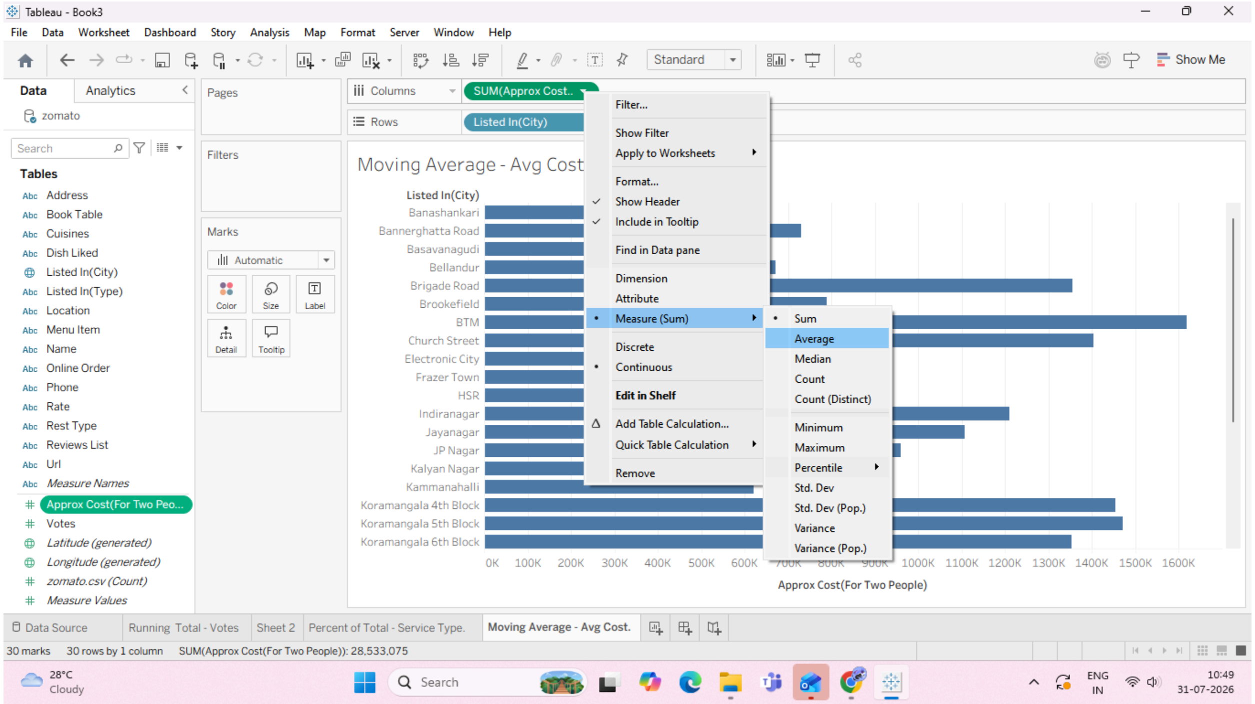
Purvi.R.Karkera
T086

B: Create moving averages and cumulative metrics Moving averages are used to smooth out short-term fluctuations and highlight longer-term trends or cycles in data.

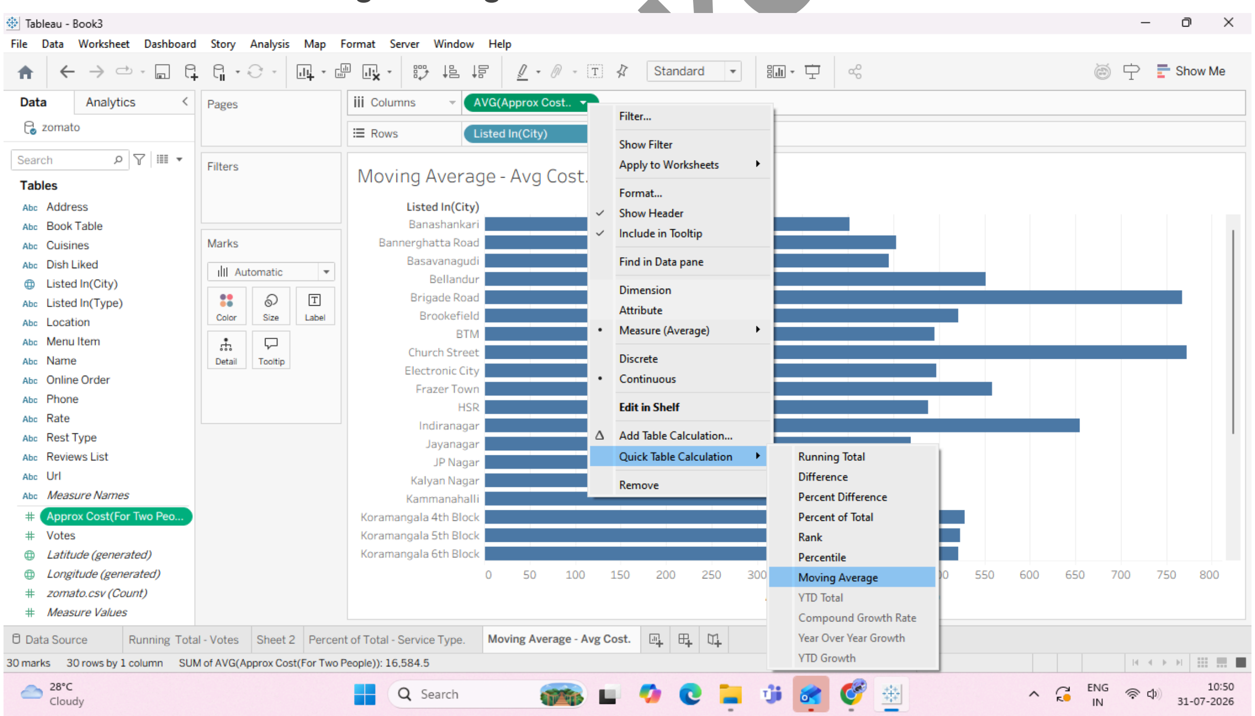
Task 1: Create a Moving Average Calculation We will create a moving average to smooth out variations in pricing across restaurants based on vote counts or listing order.



SHETH L.U.J & SIR MV COLLEGE



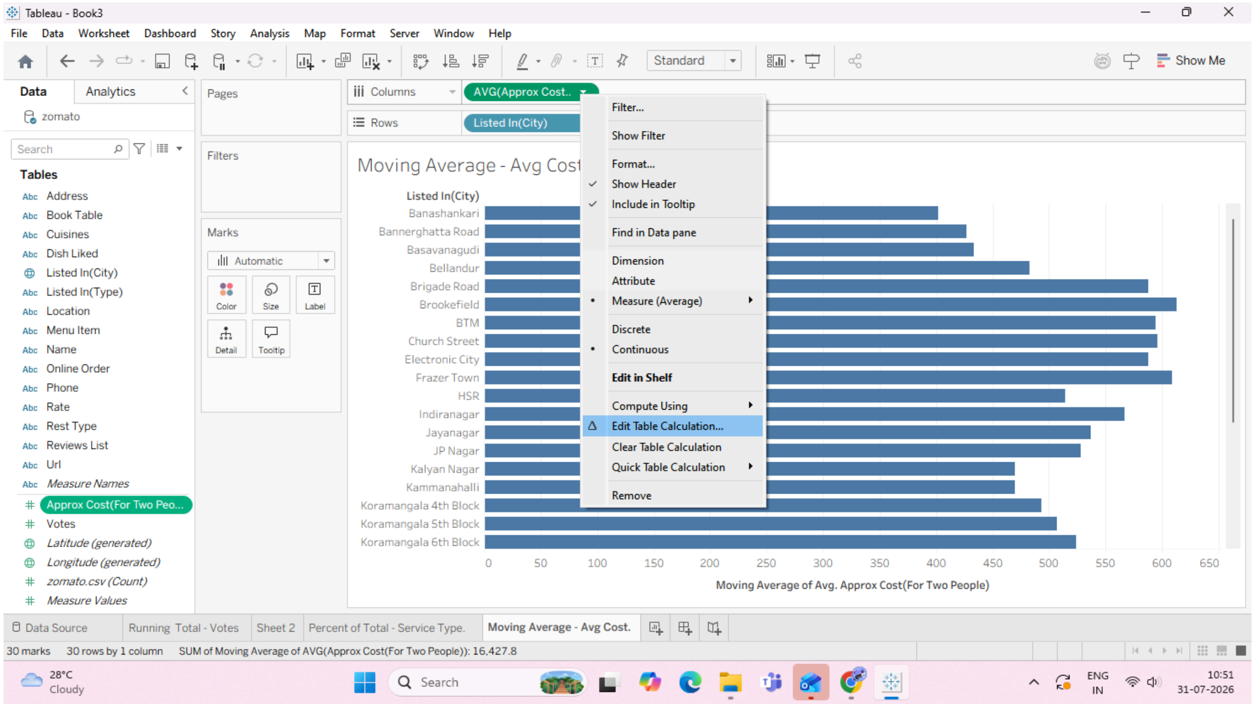
4. Convert to a Moving Average Quick Table Calculation:

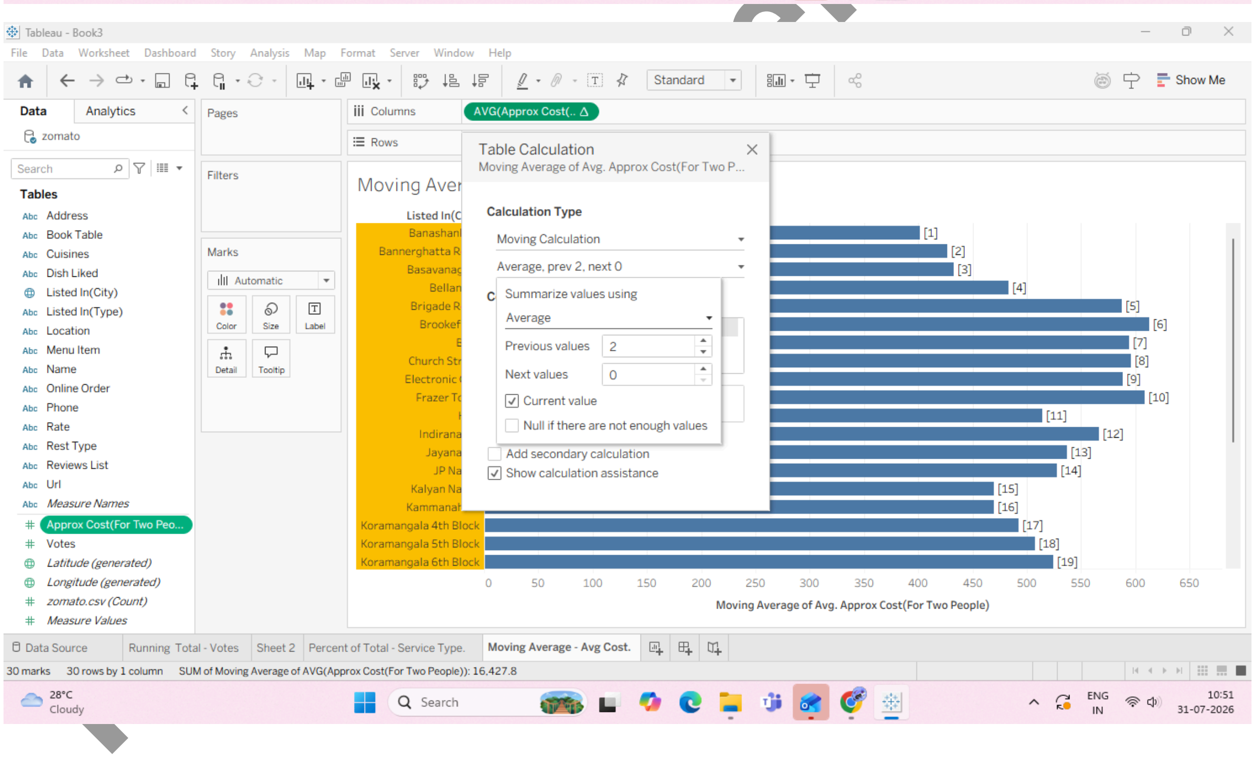


5. Customize the Moving Window:

Purvi.R.Karkera
T086

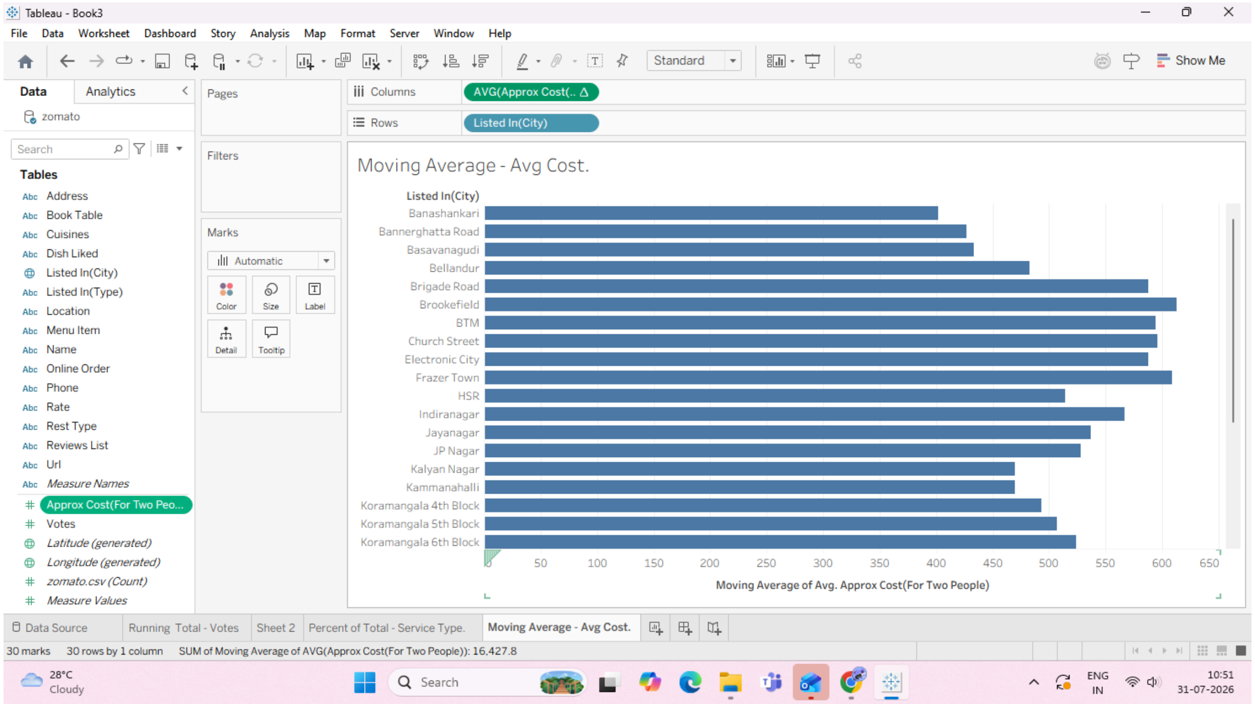
SHETH L.U.J & SIR MV COLLEGE



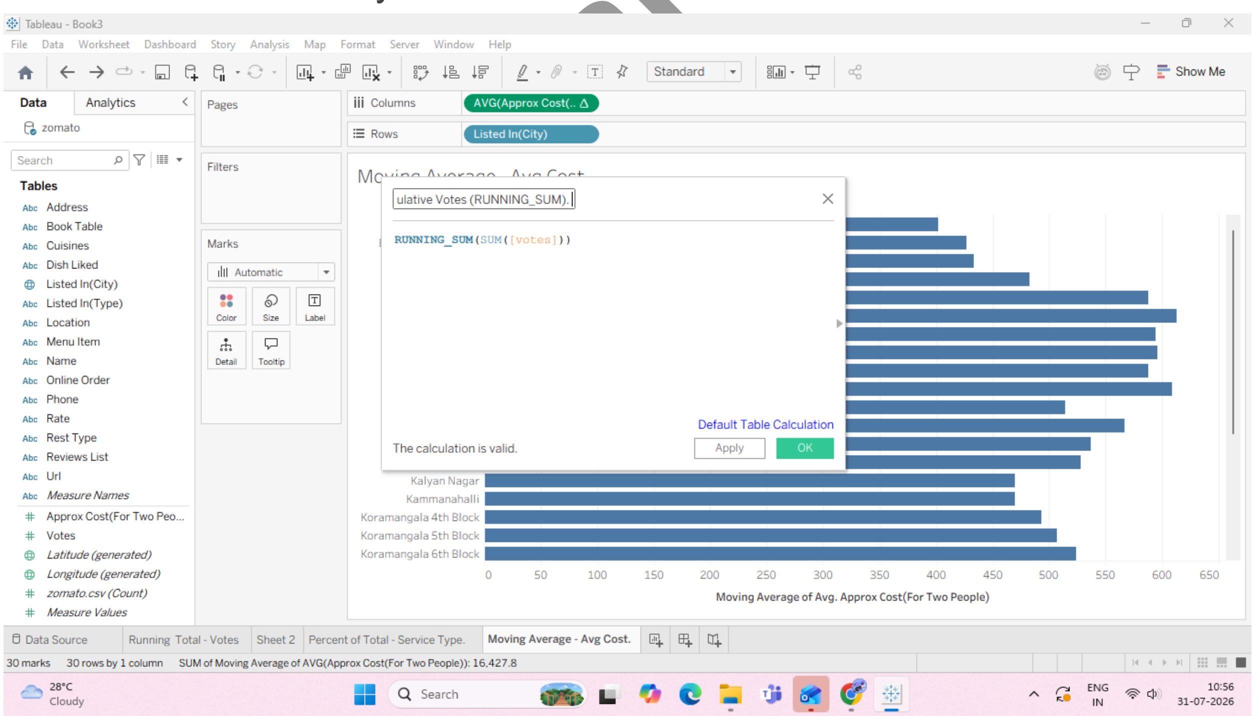


Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE



Task 2: Create a Custom Cumulative Metric via Calculated Field Instead of using Quick Table Calculations, we can write a dedicated Table Calculation function formula directly.



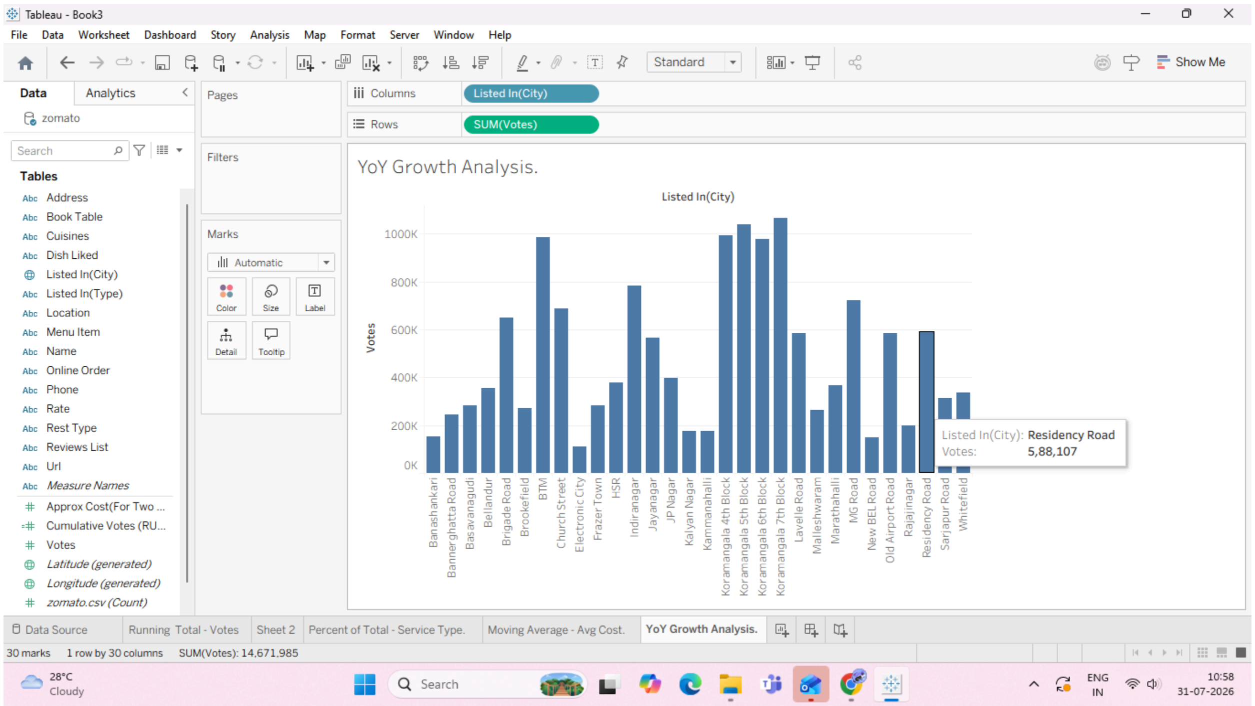
Purvi.R.Karkera
T086

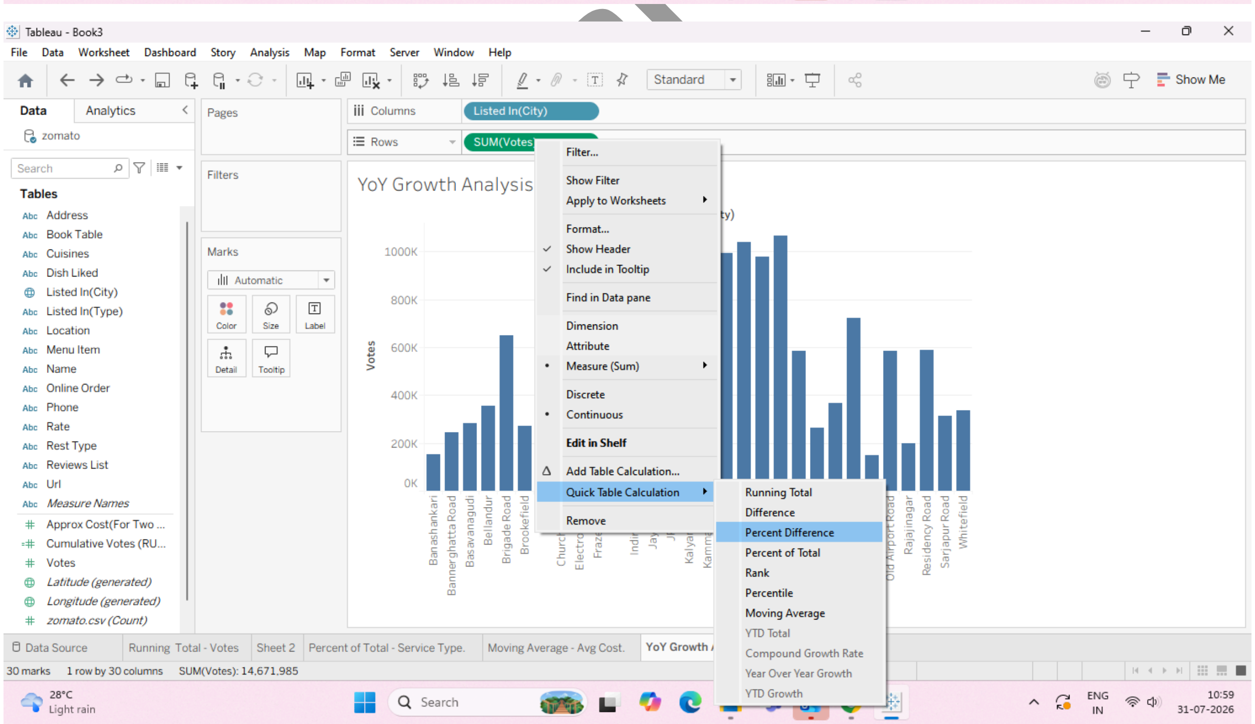
SHETH L.U.J & SIR MV COLLEGE

C: Perform year-over-year growth analysis

Task: Calculate Year-over-Year (YoY) / Period-over-Period Percent Growth

4. Apply the Year-over-Year Growth Quick Table Calculation:

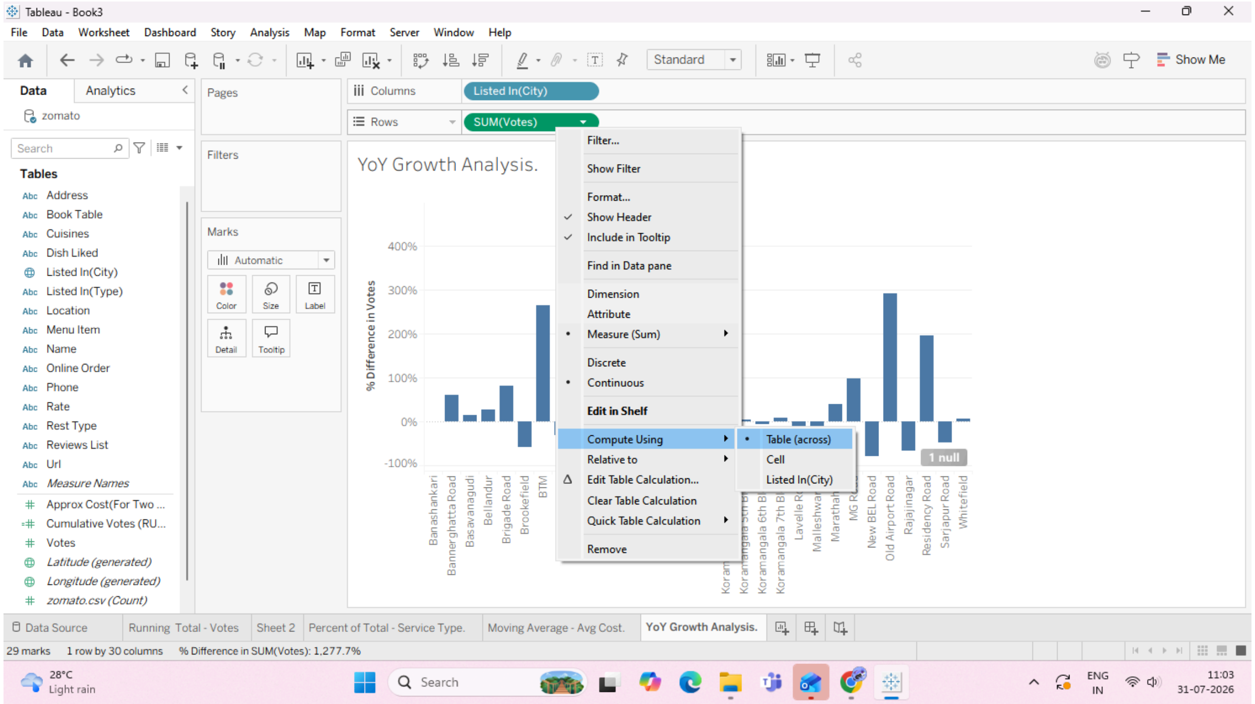




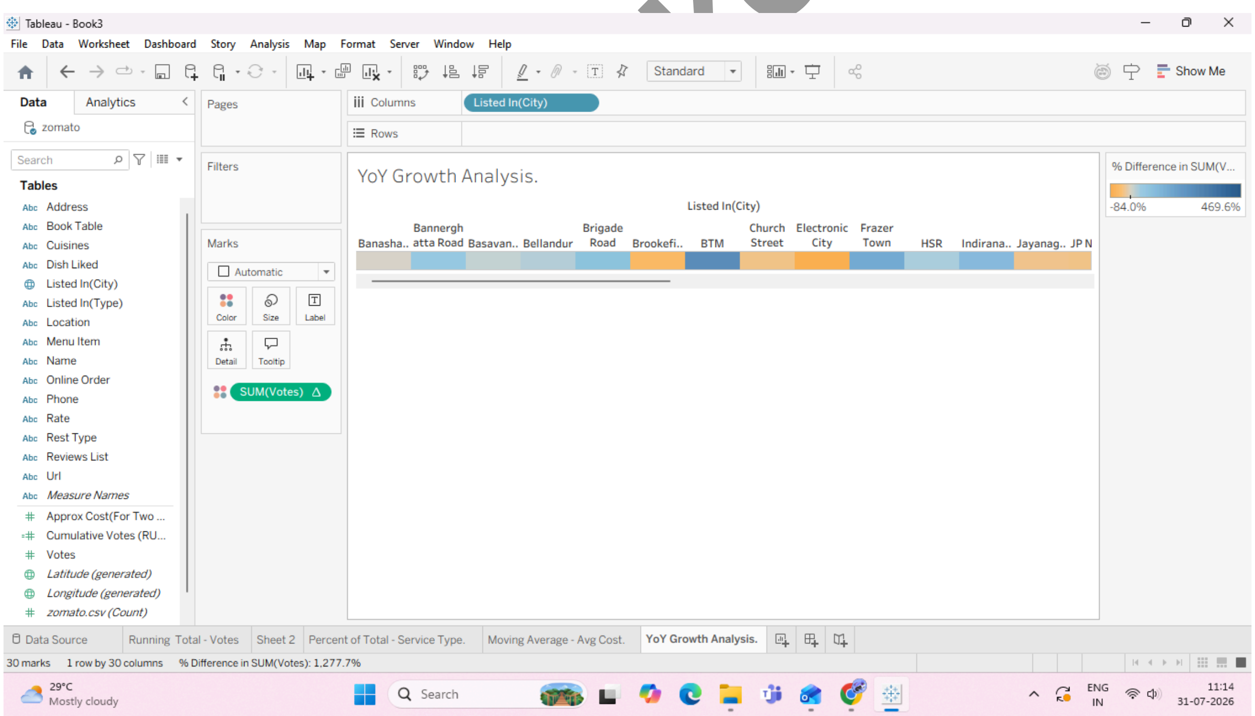
5. Configure the Growth Calculation Direction:

Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE



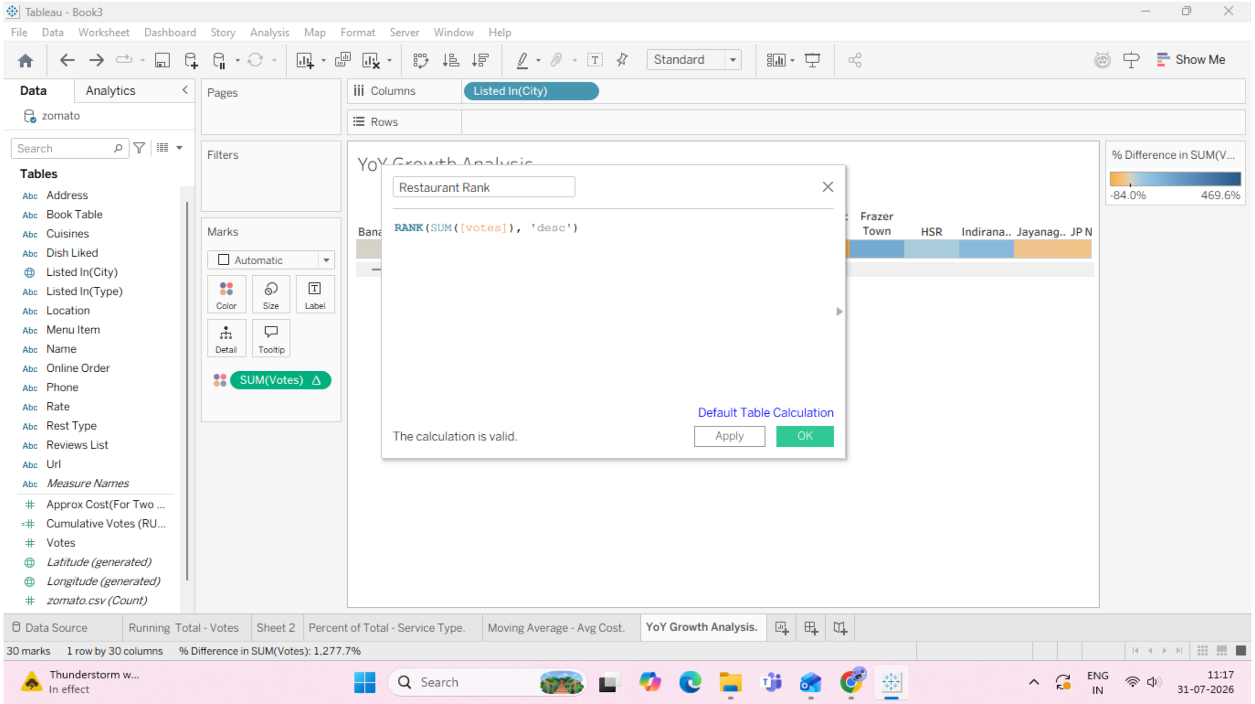
6. Add Visual Color Indicators:



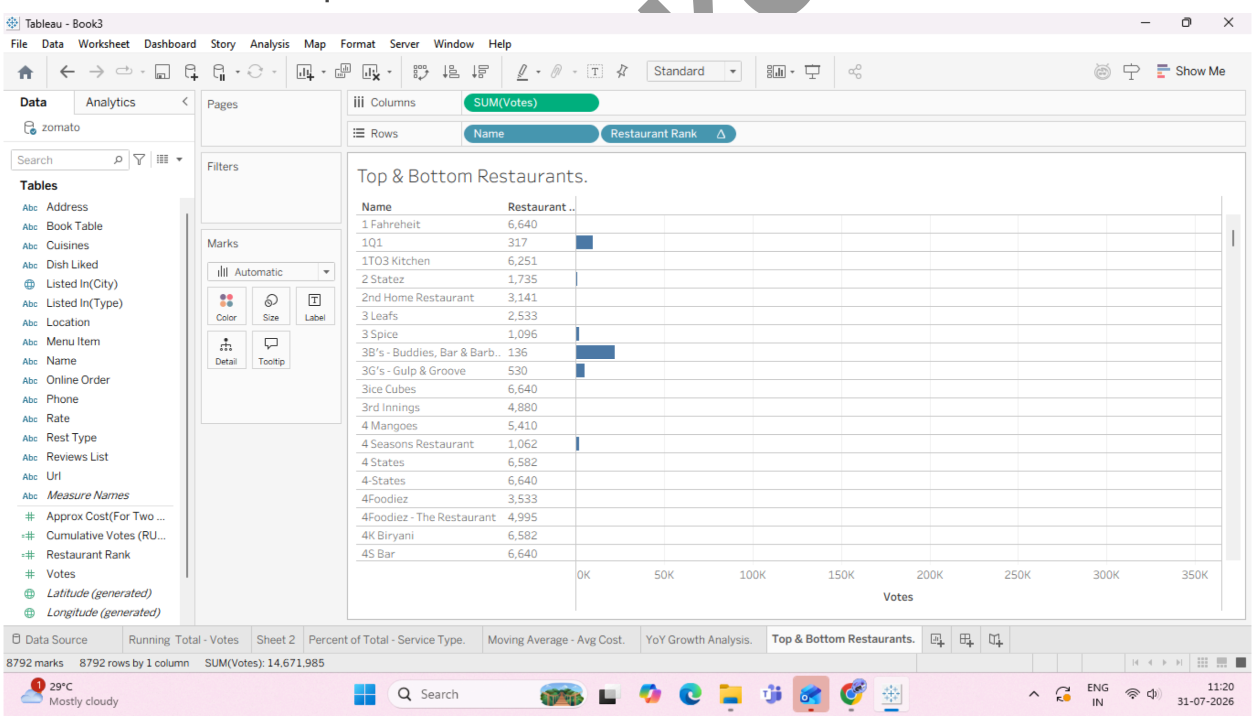
D: Use ranking functions (Top N, Bottom N) Ranking table calculations evaluate each row's aggregate value relative to all other rows in the partition, assigning a numeric rank (1, 2, 3...). Task 1: Create a Rank Calculation

Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

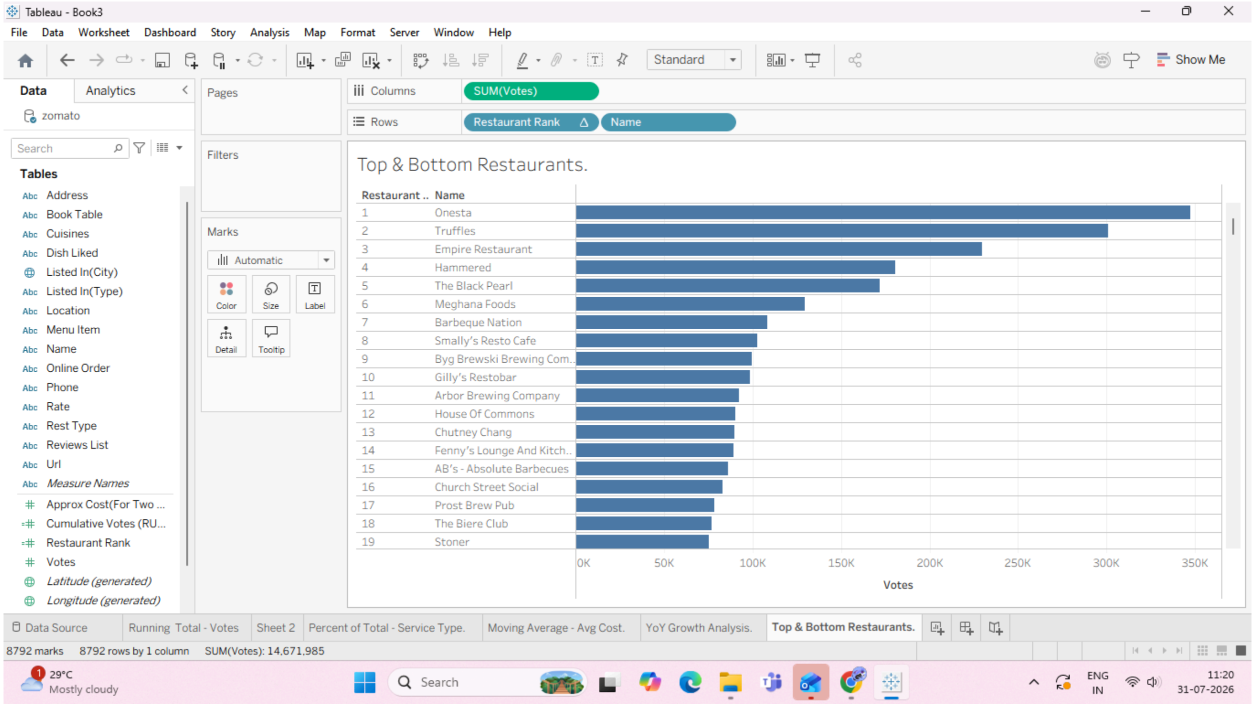


Task 2: Build the Top N / Bottom N Ranked View

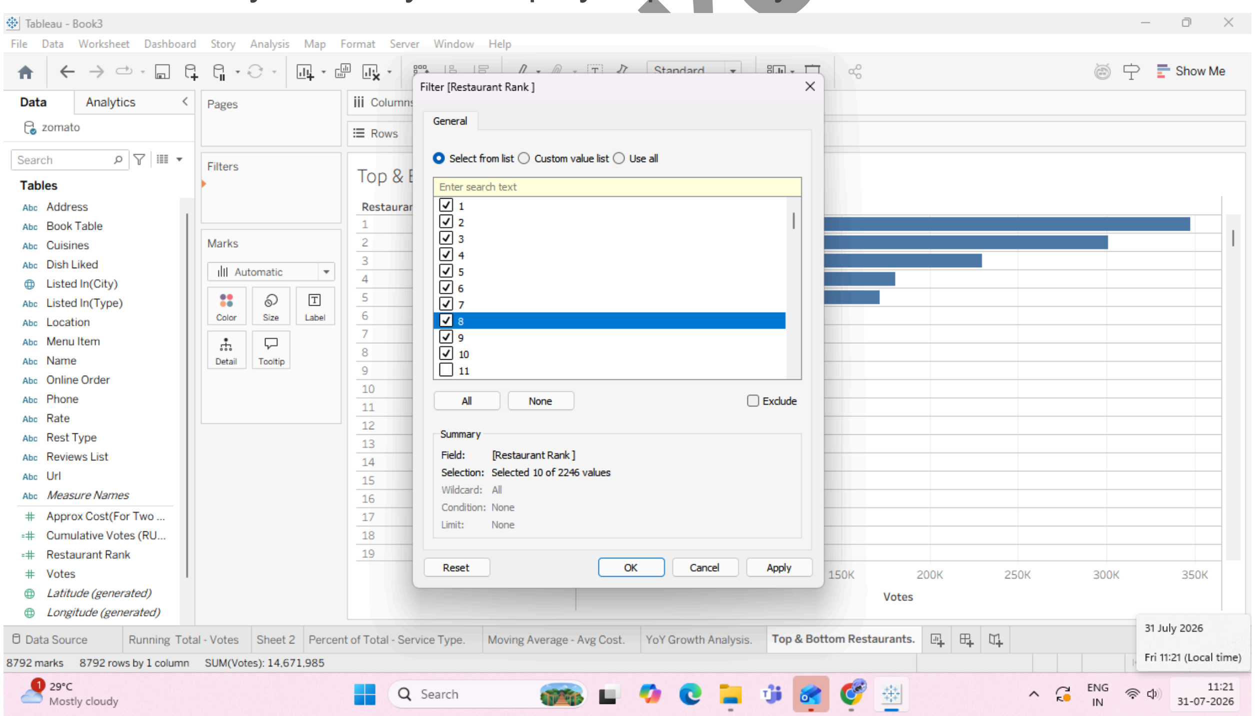


Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

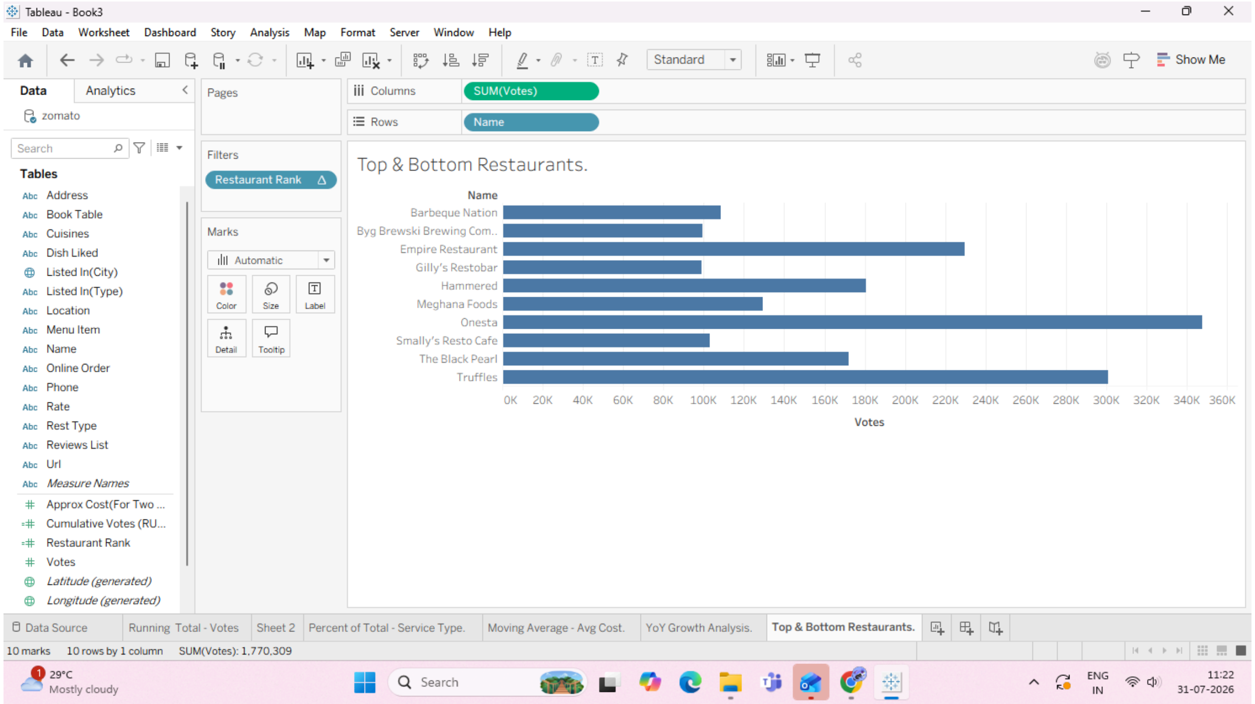


Task 3: Filter Dynamically to Display Top N Only



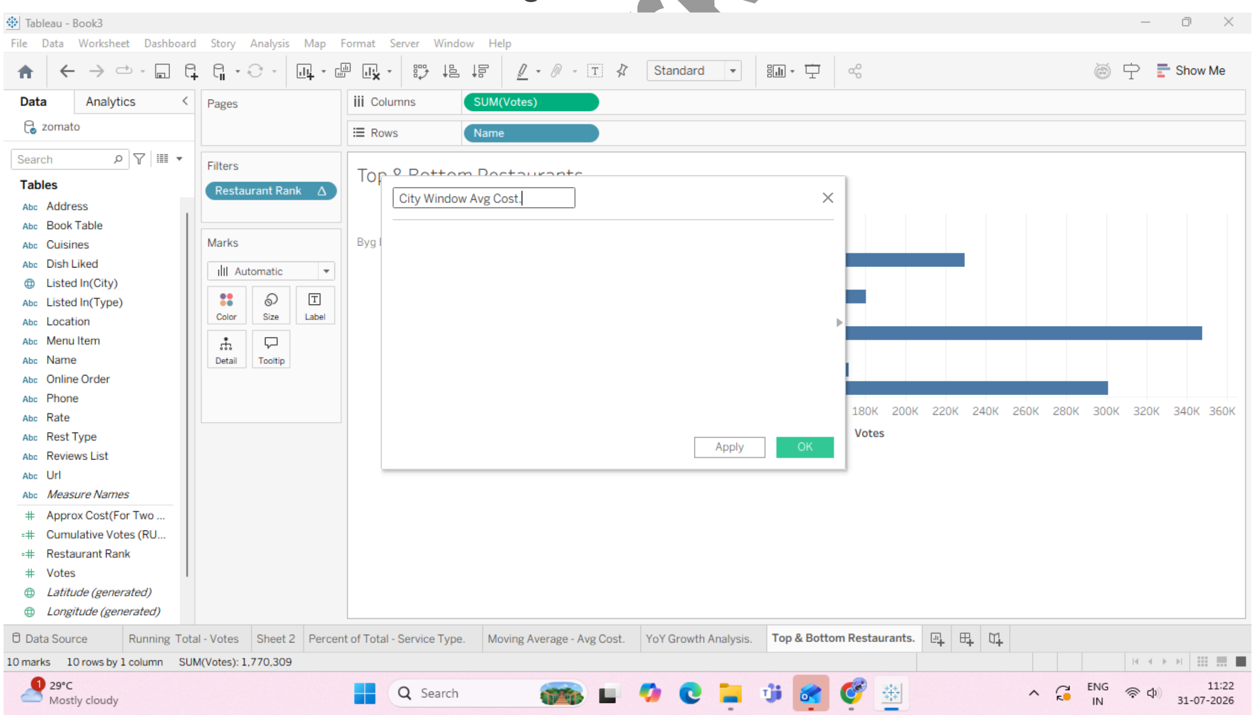
Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE



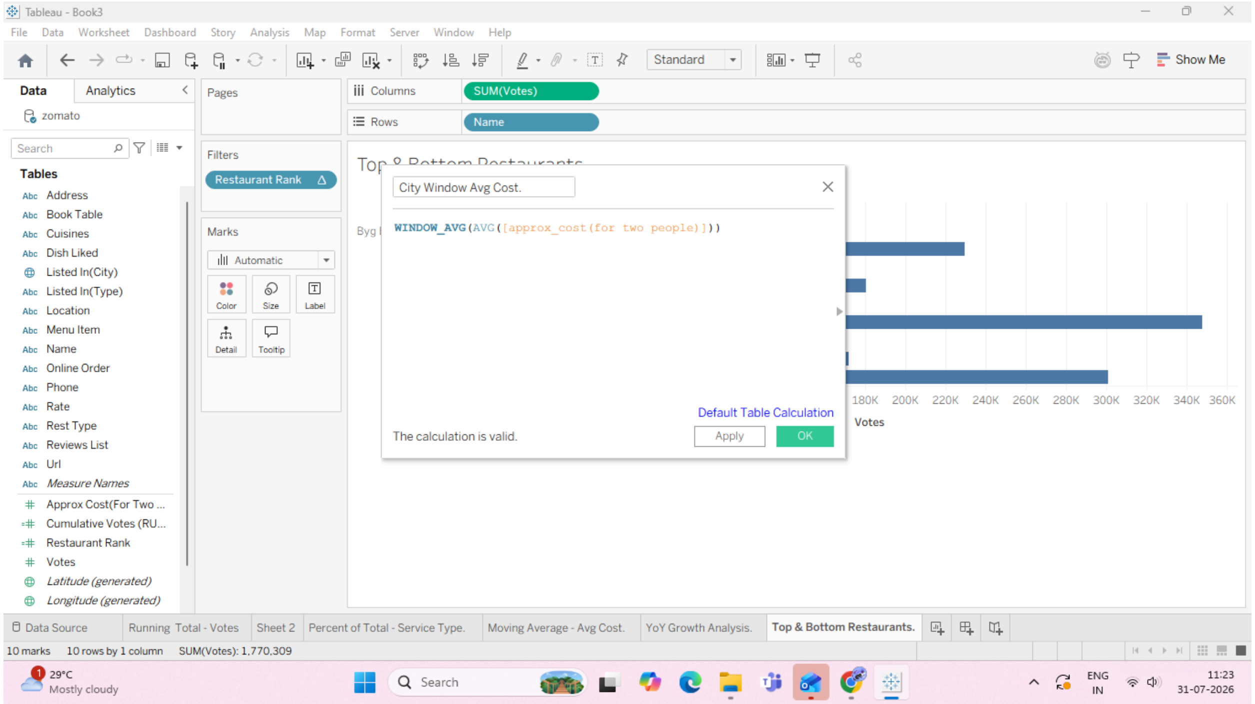
E: Implement window functions (WINDOW_SUM, WINDOW_AVG)

Task 1: Create a Window Average Calculation

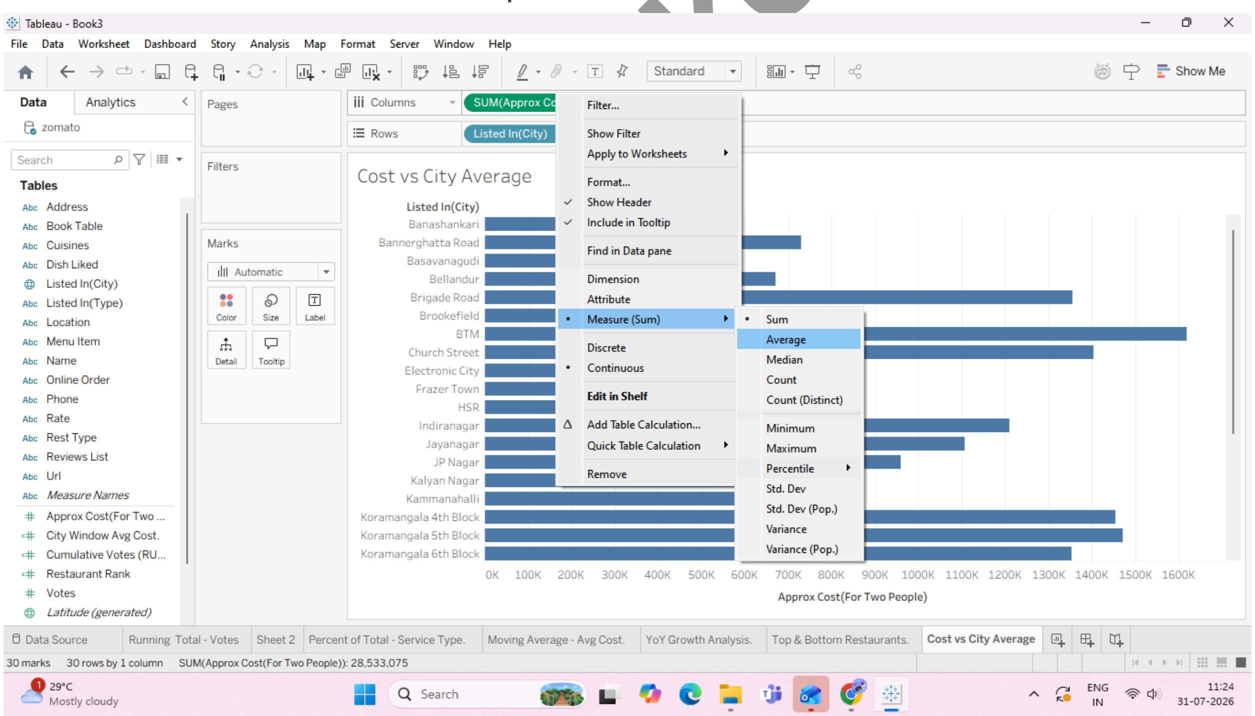


Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

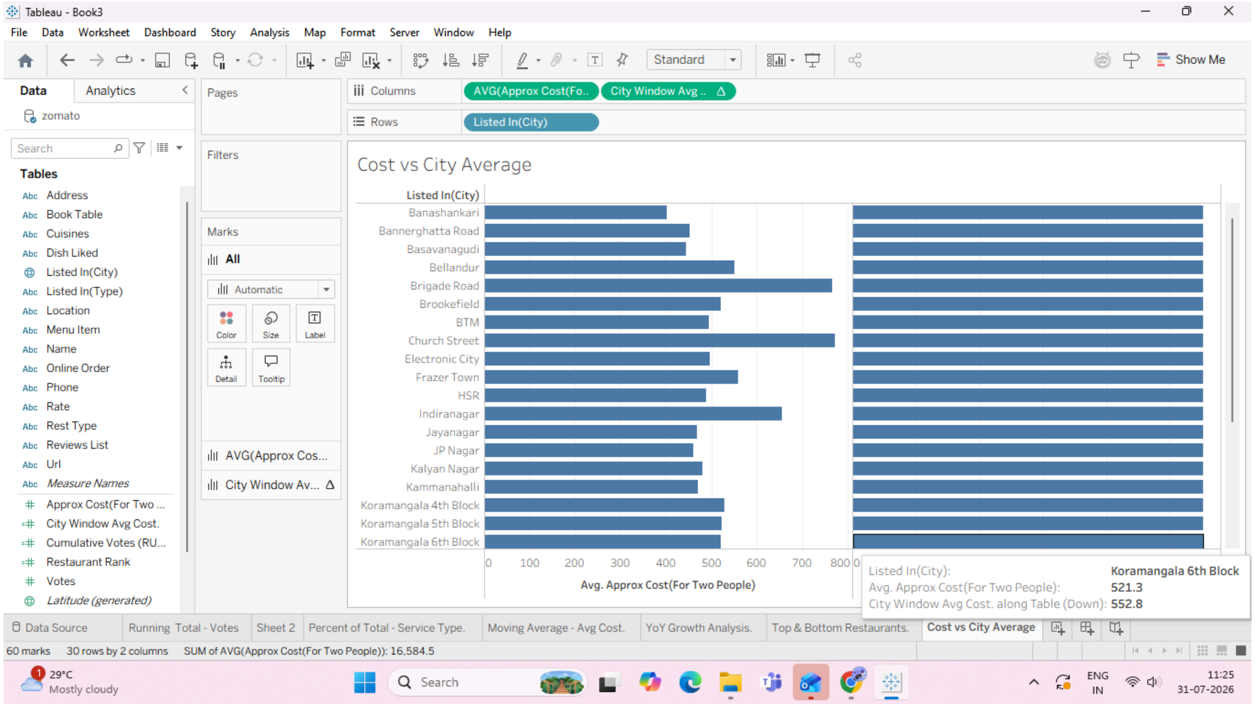


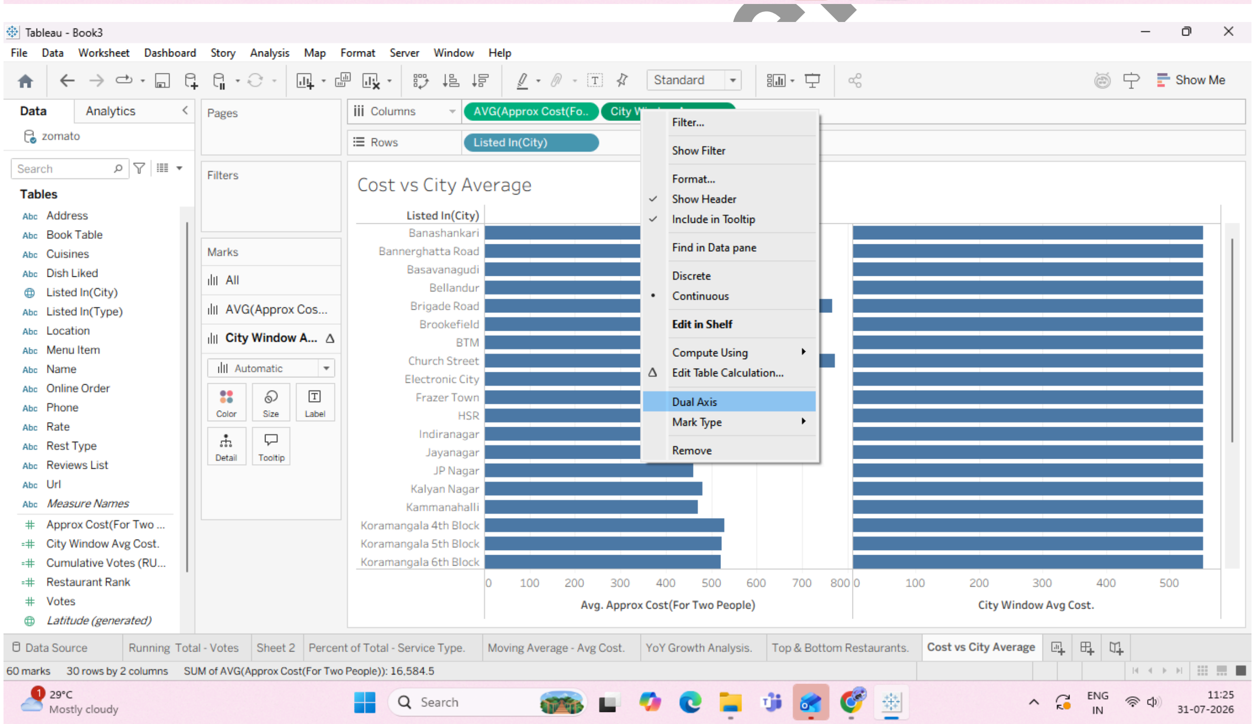
Task 2: Build the Window Comparison Visualization



Purvi.R.Karkera
T086

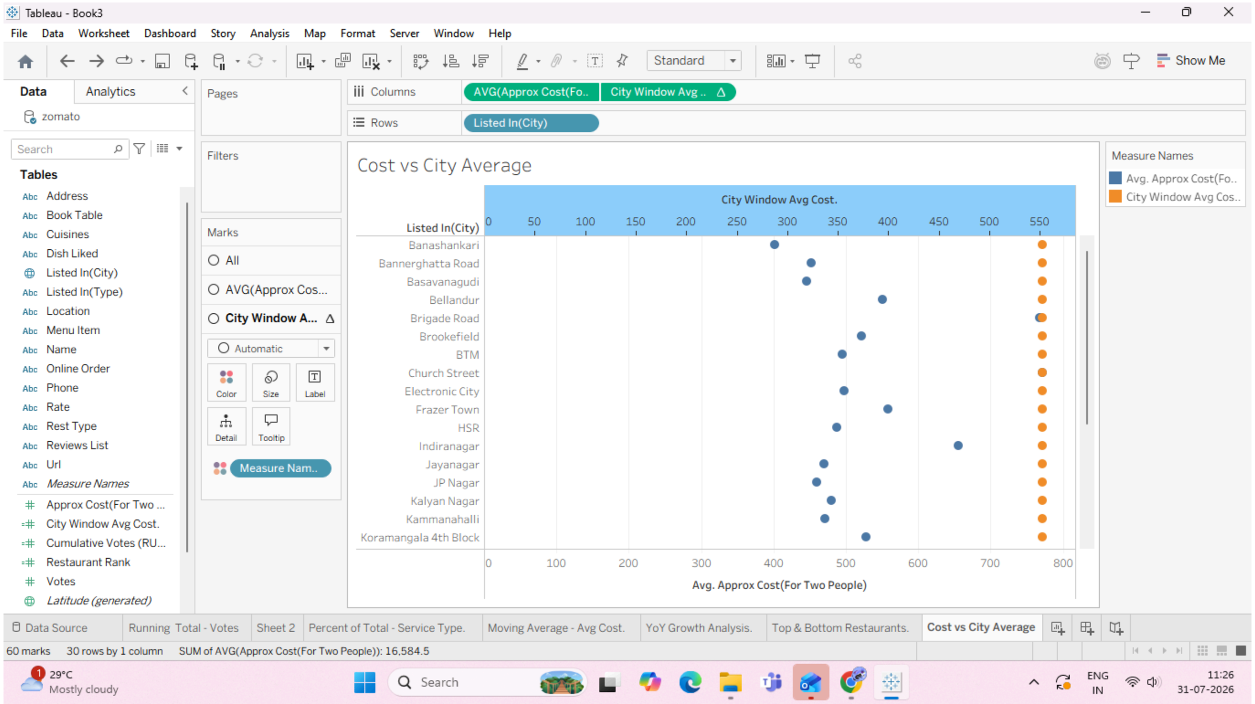
SHETH L.U.J & SIR MV COLLEGE

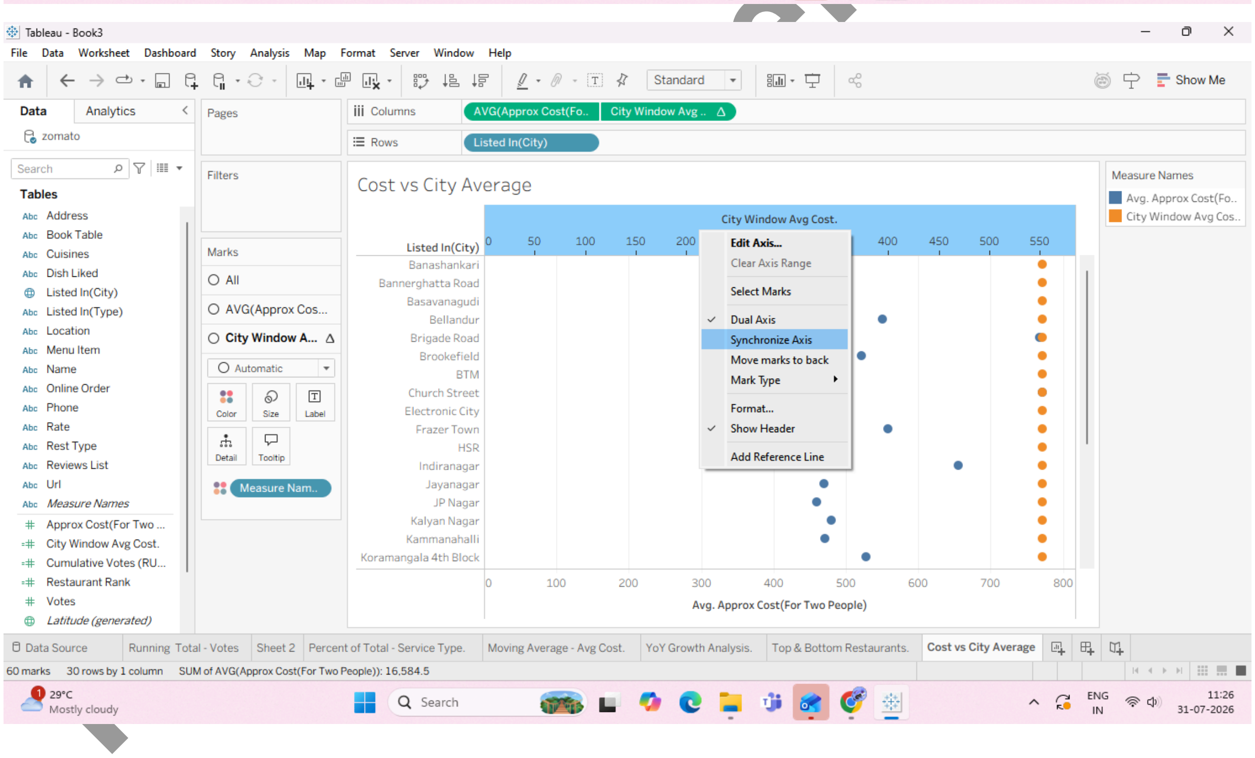




Purvi.R.Karkera
T086

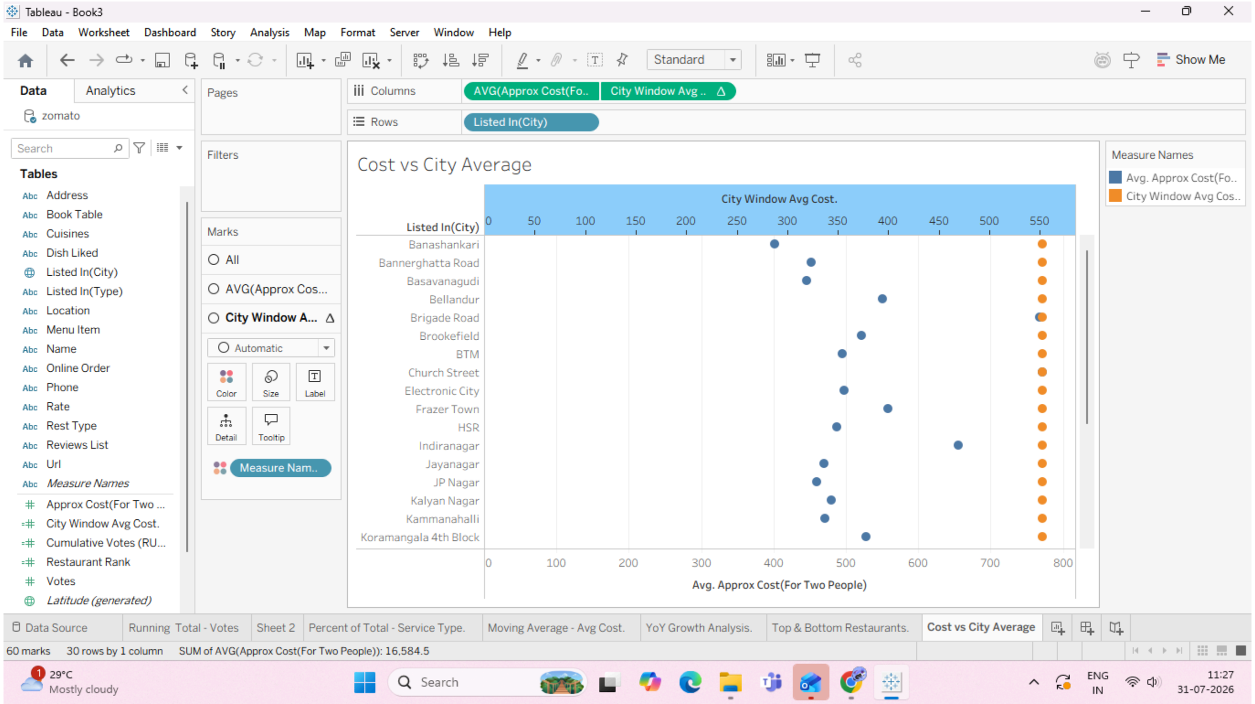
SHETH L.U.J & SIR MV COLLEGE

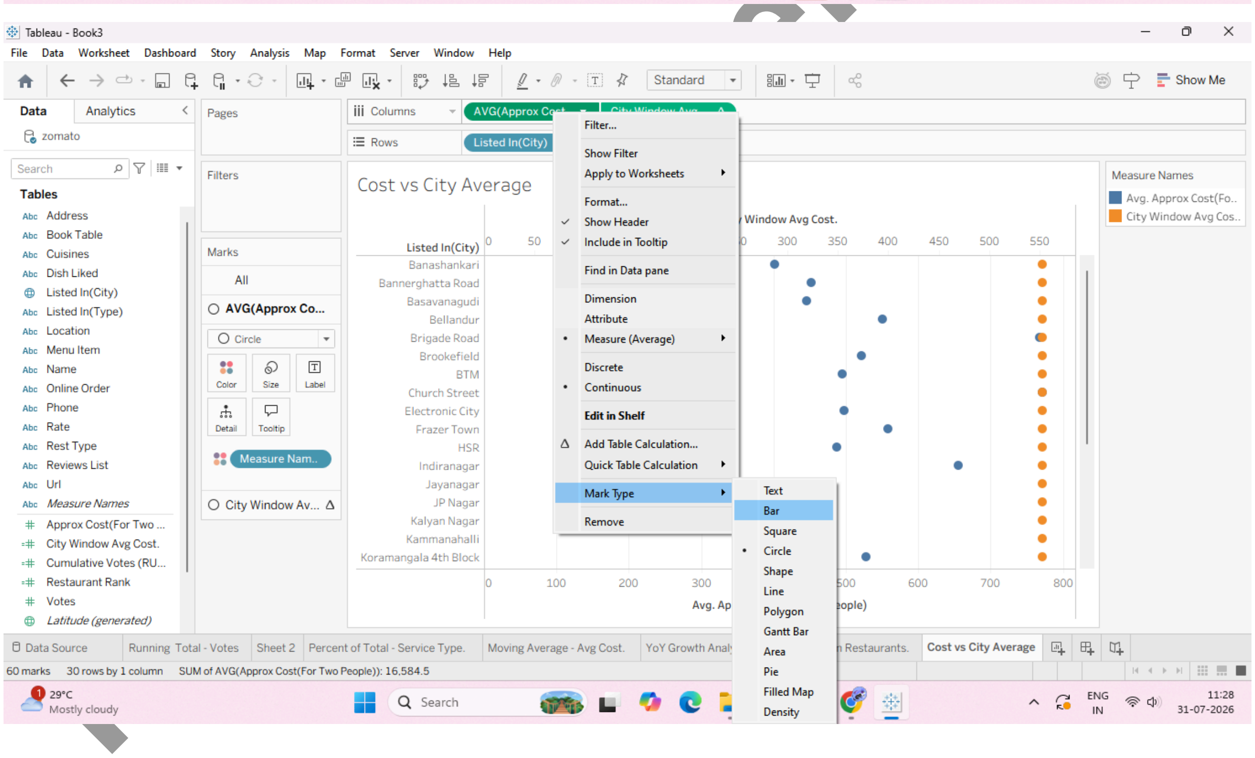




Purvi.R.Karkera
T086

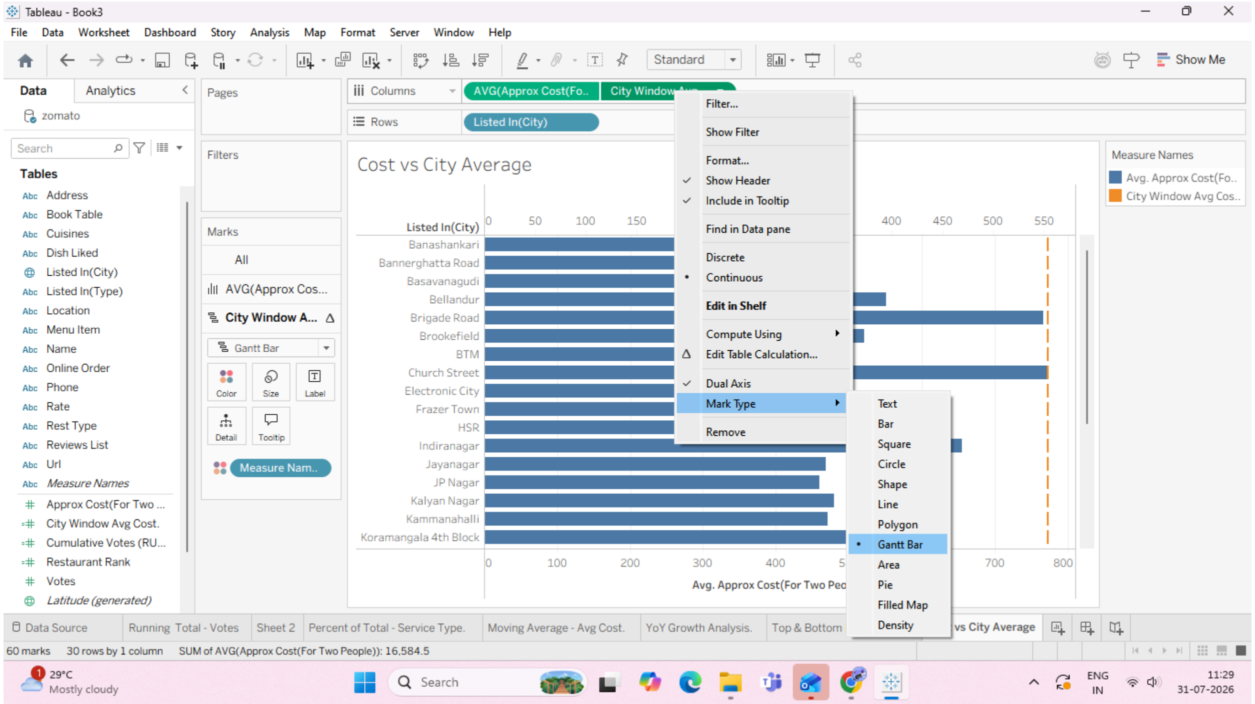
SHETH L.U.J & SIR MV COLLEGE



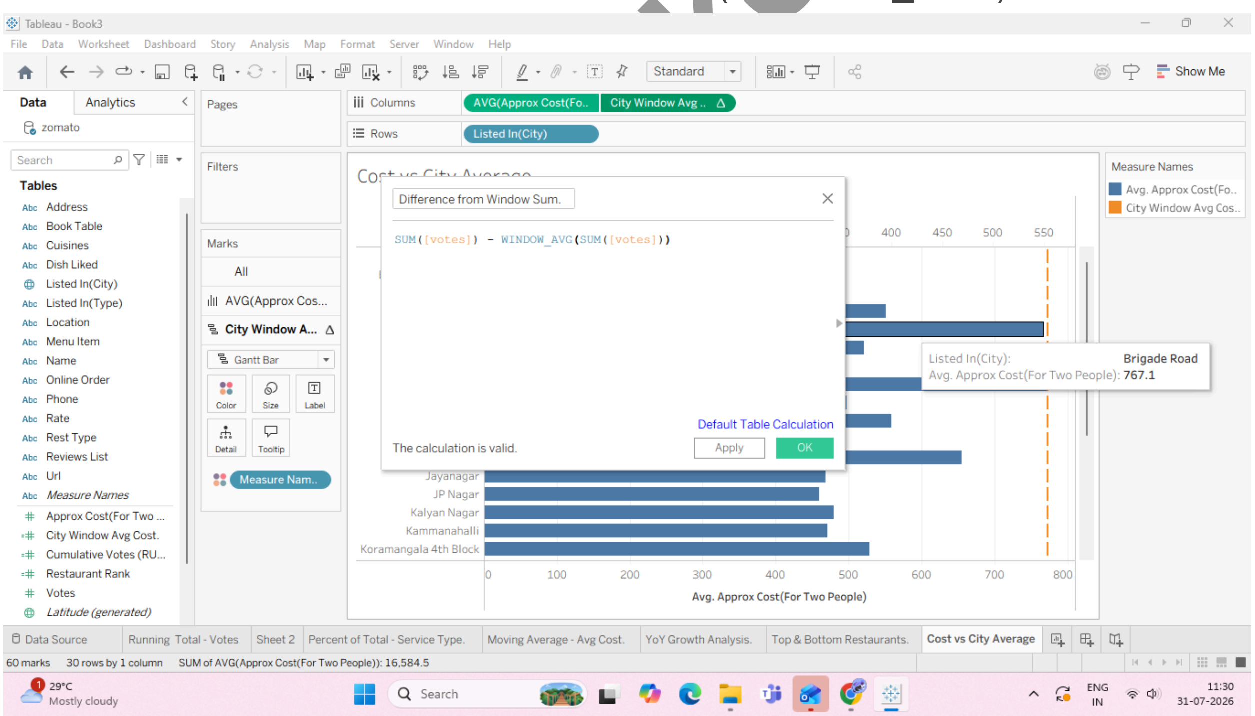


Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE



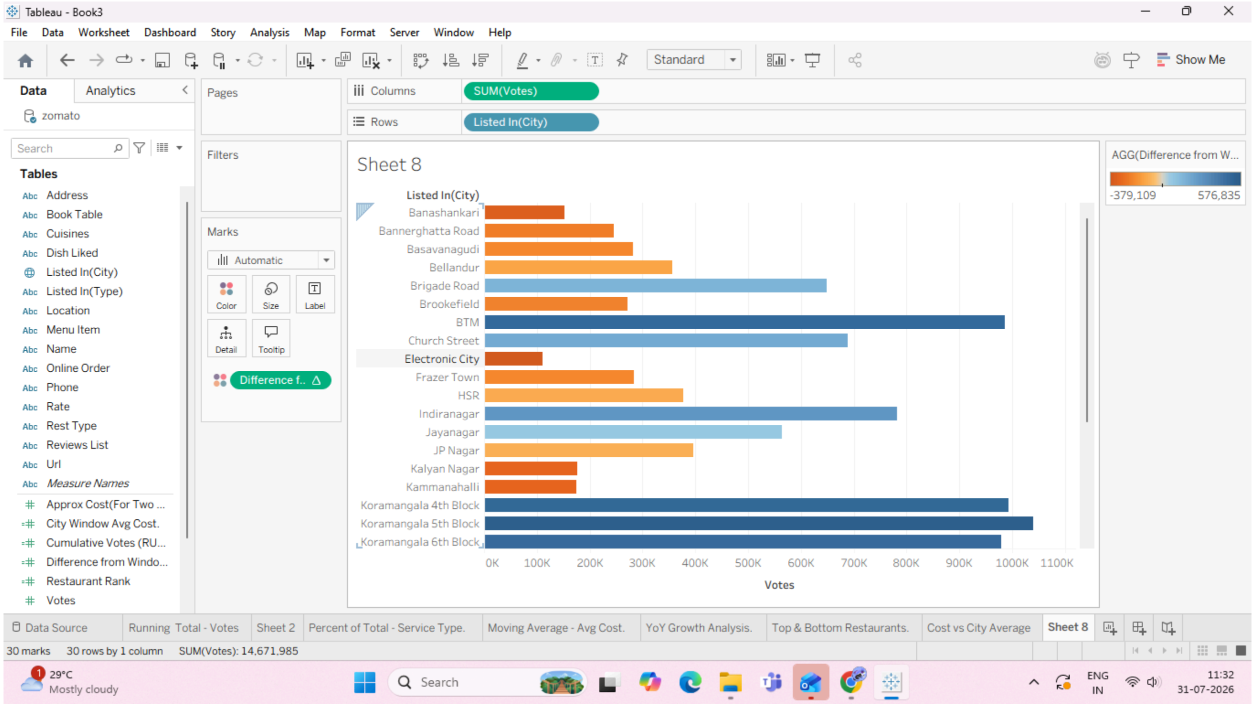
Task 4: Create a Window Difference Metric (WINDOW_SUM)

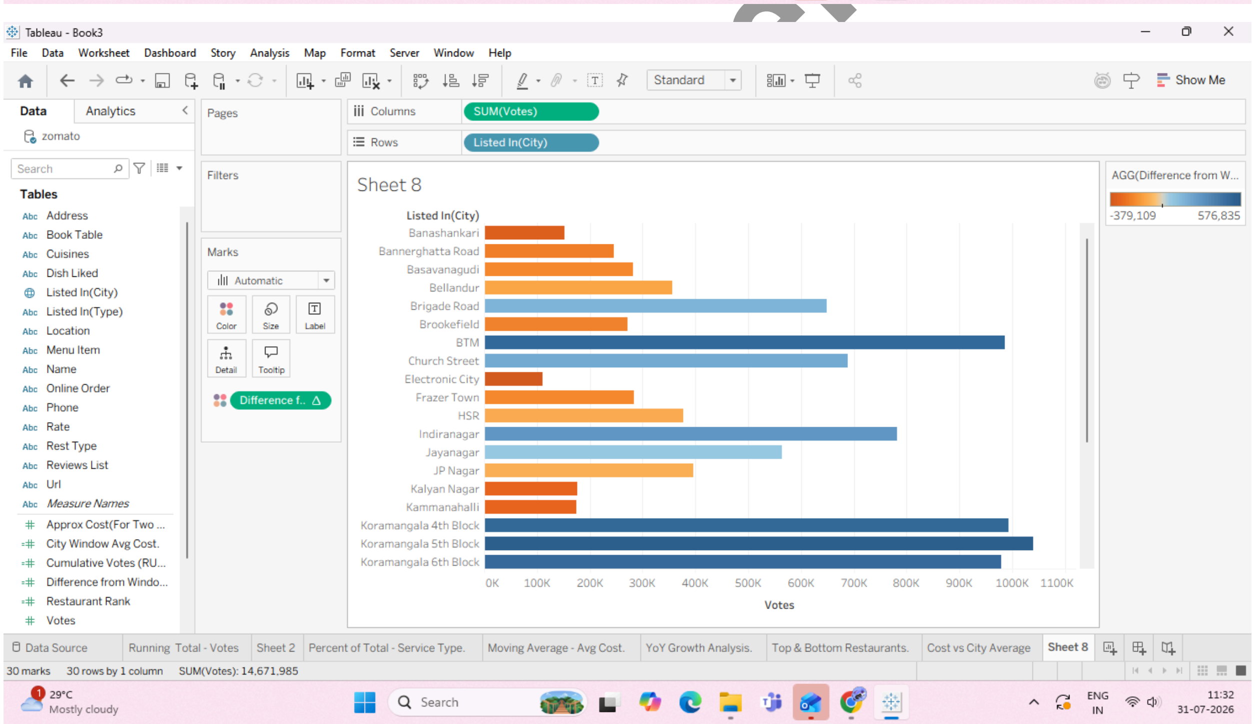


5. Apply it to the View:

Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

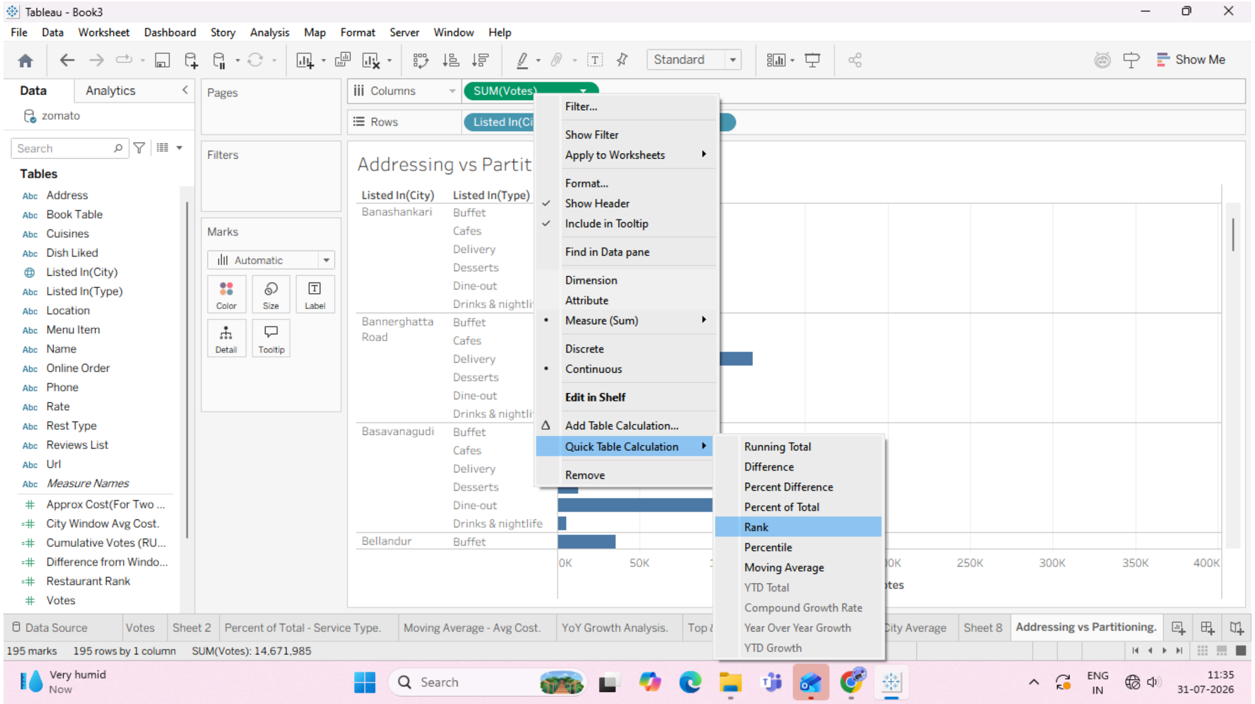


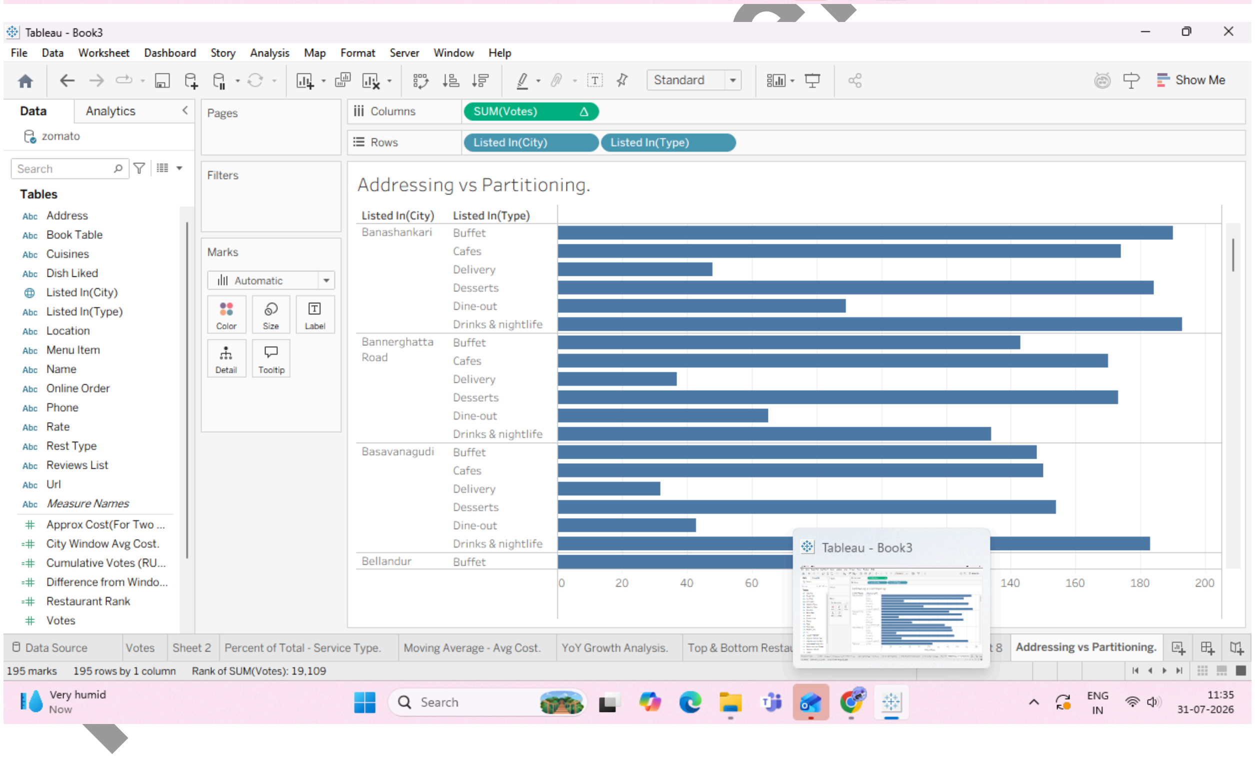


F: Customize addressing and partitioning

Purvi.R.Karkera
T086

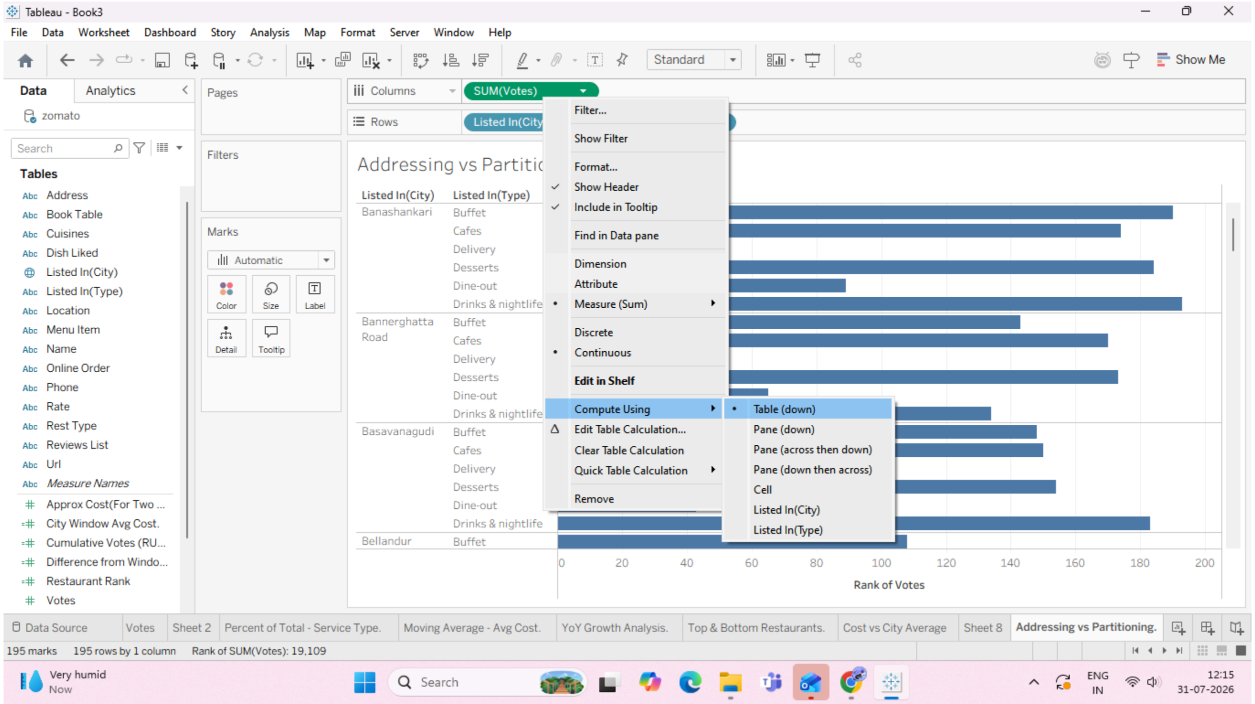
SHETH L.U.J & SIR MV COLLEGE

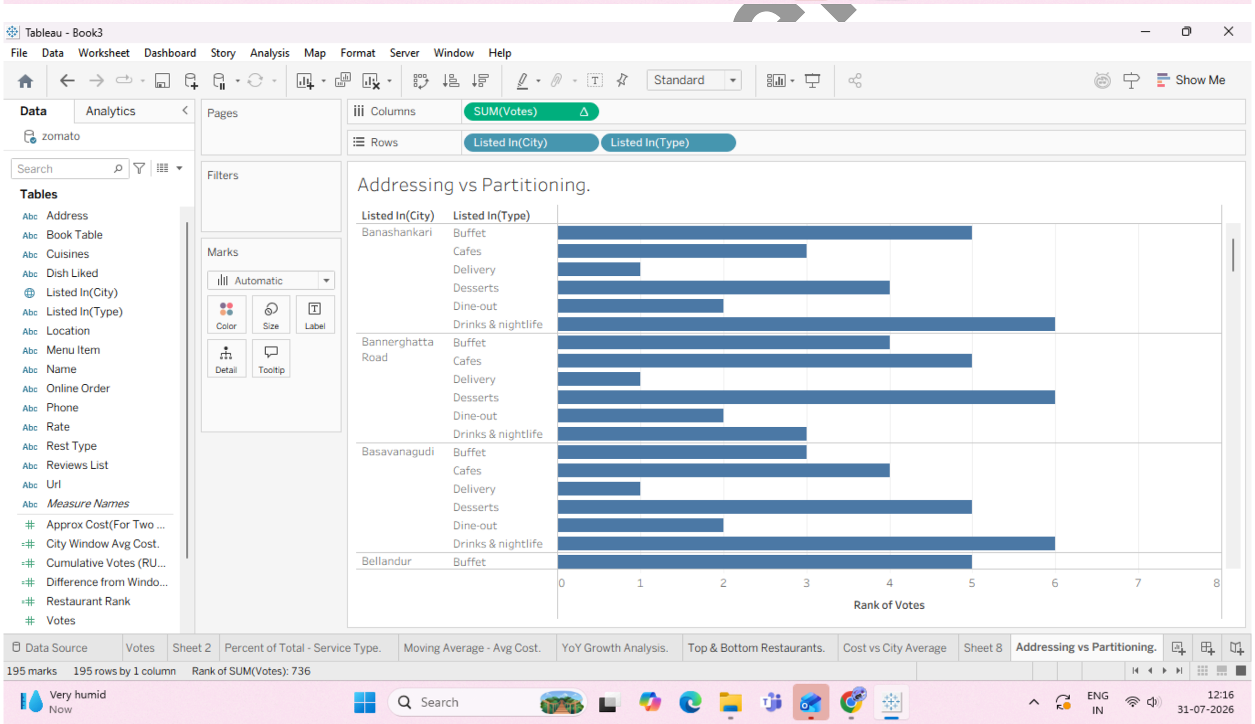




Purvi.R.Karkera
T086

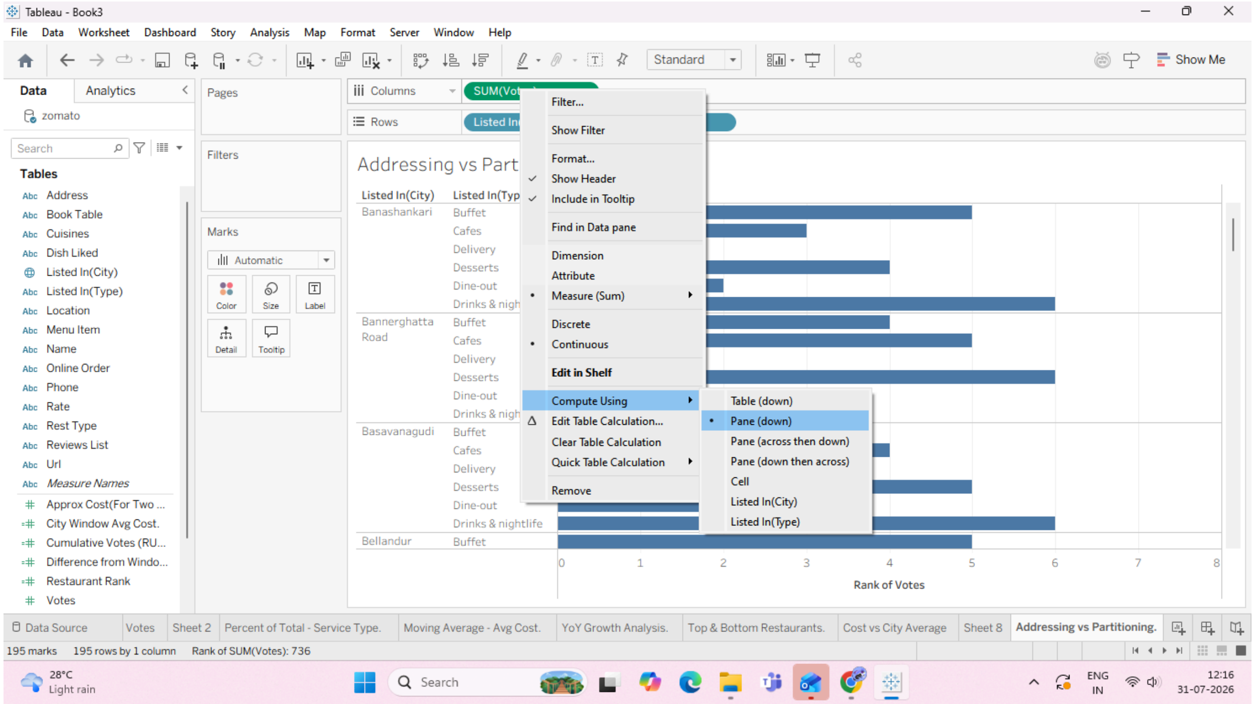
SHETH L.U.J & SIR MV COLLEGE

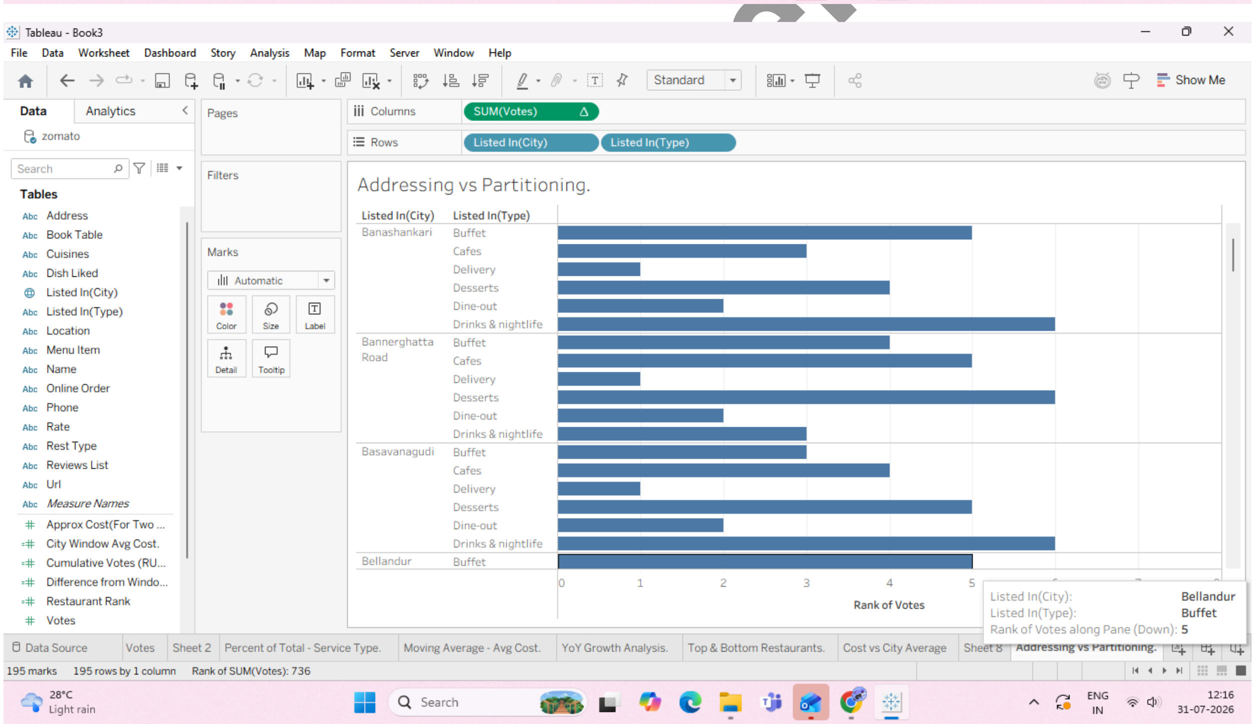




Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

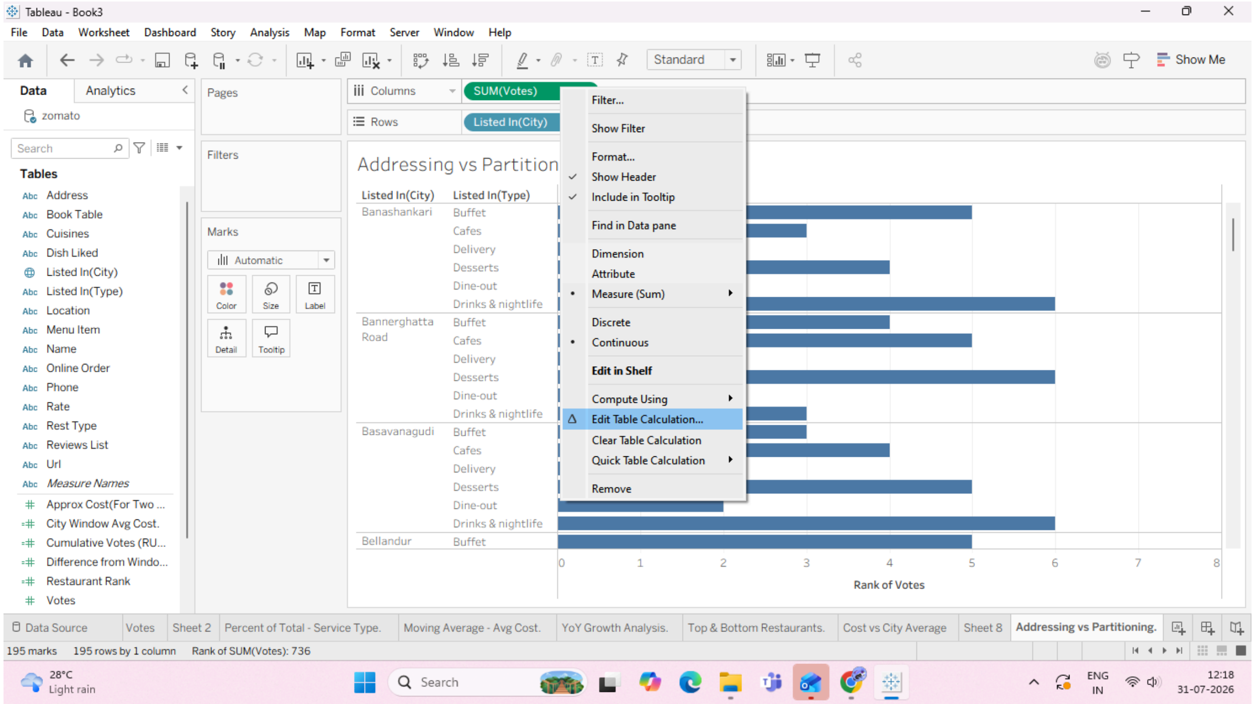


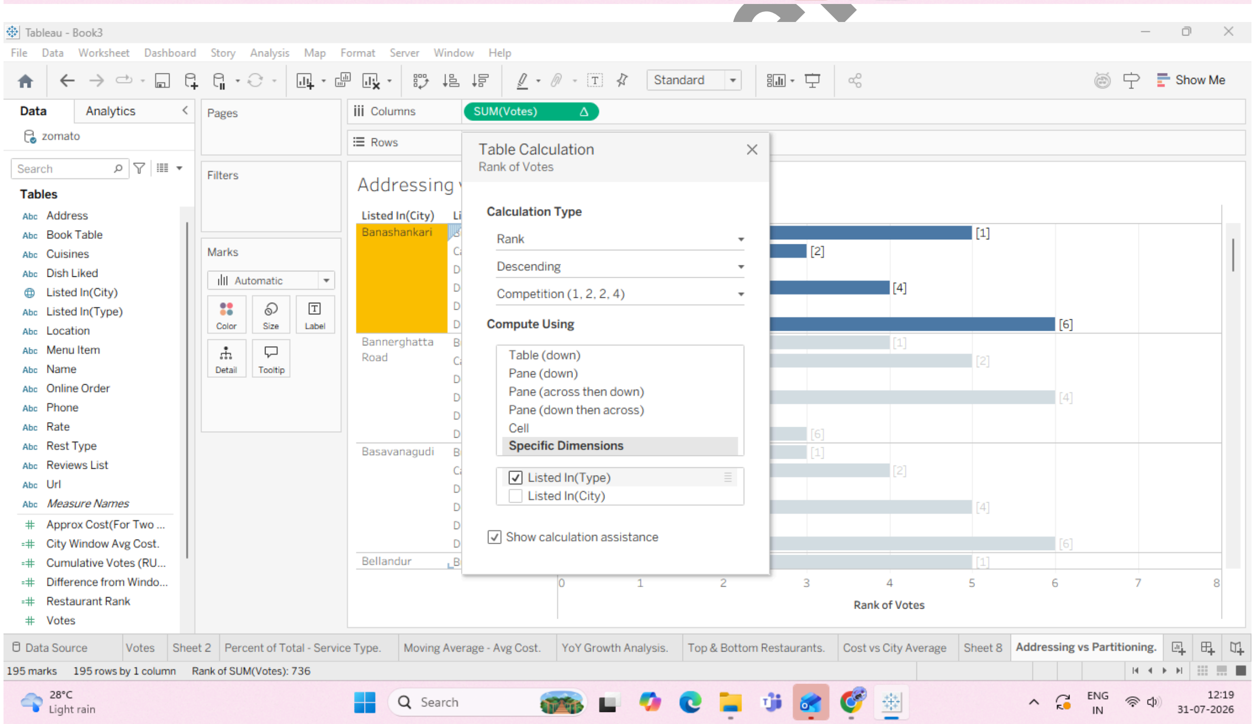


Task 3: Manually Define Addressing with Specific Dimensions

Purvi.R.Karkera
T086

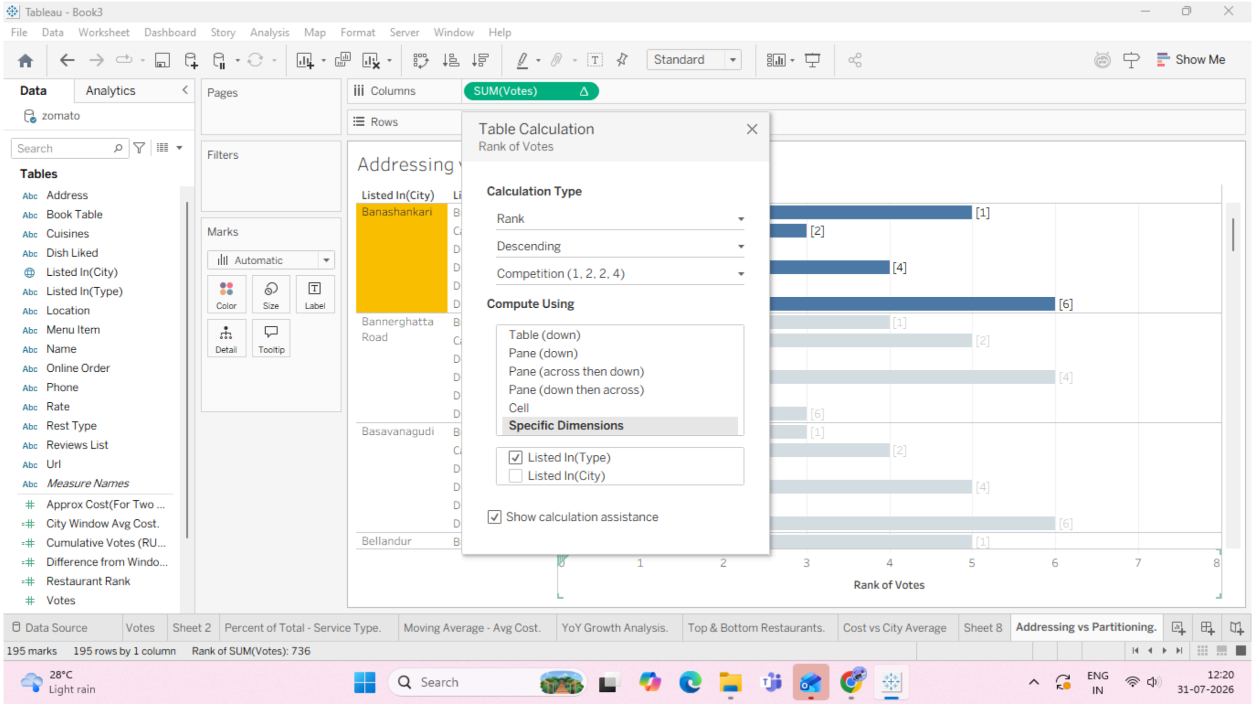
SHETH L.U.J & SIR MV COLLEGE

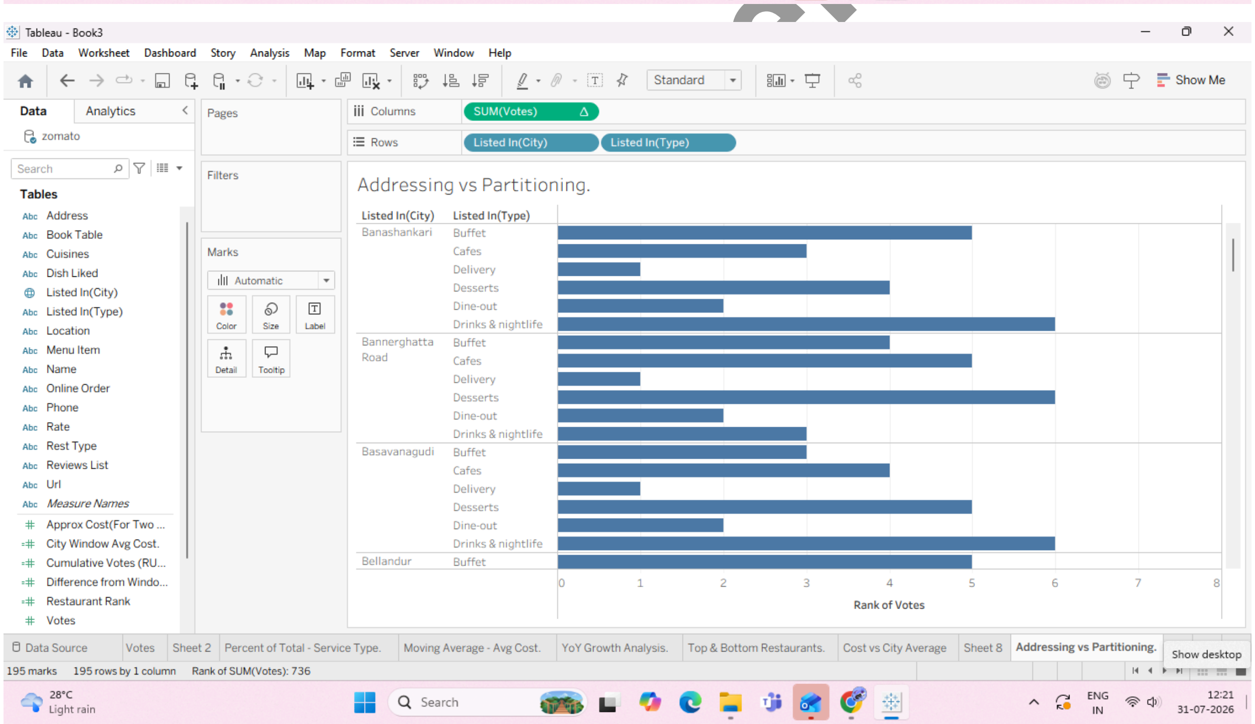




Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE



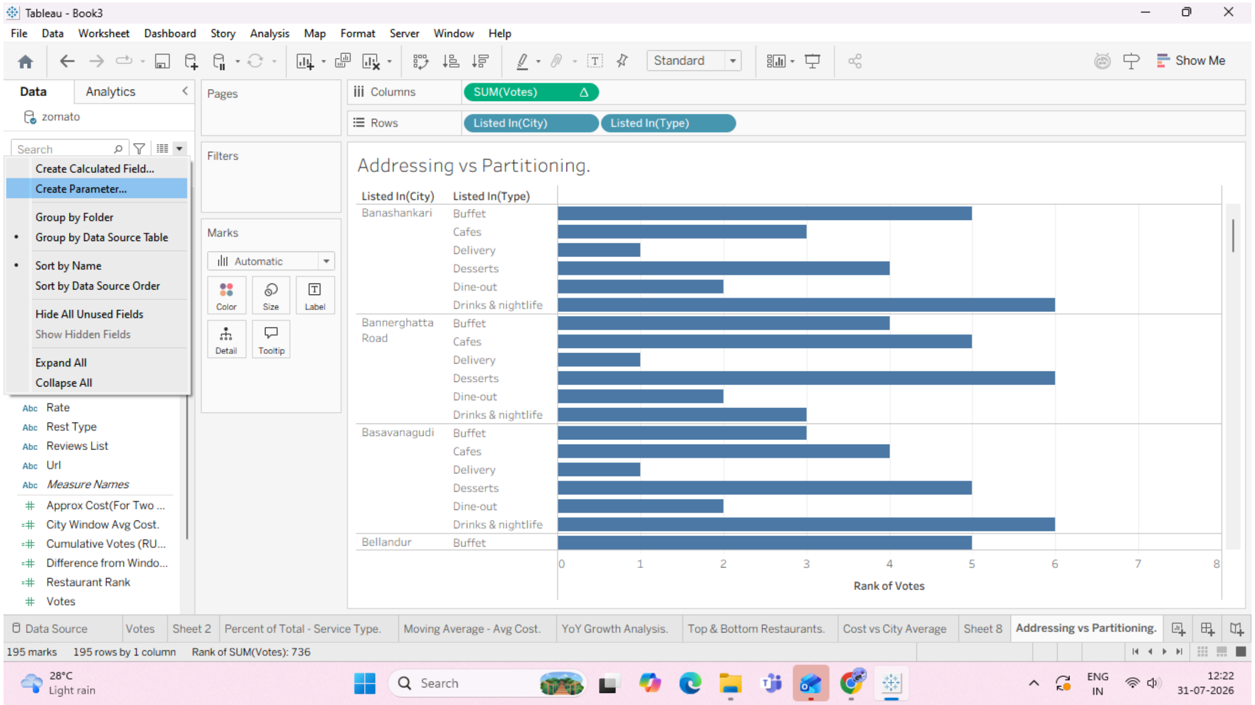


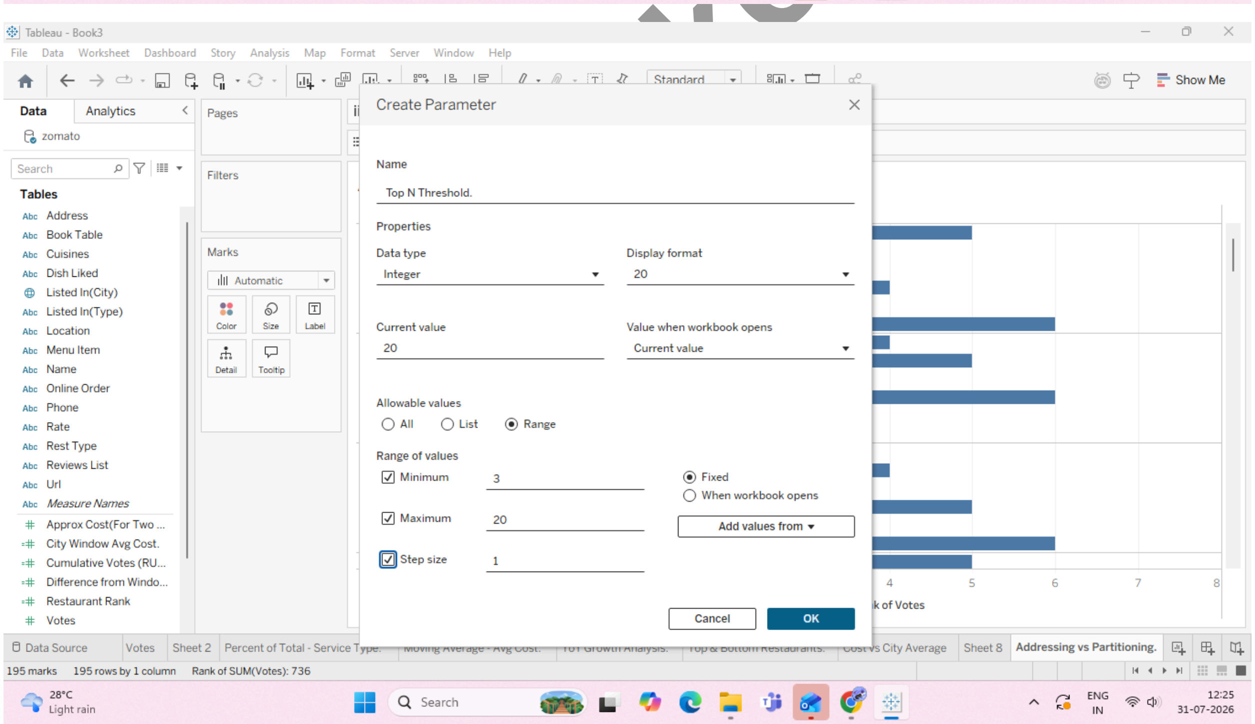
G: Create dynamic titles and labels using table calculations

Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

Task 1: Create a Dynamic Label Parameter

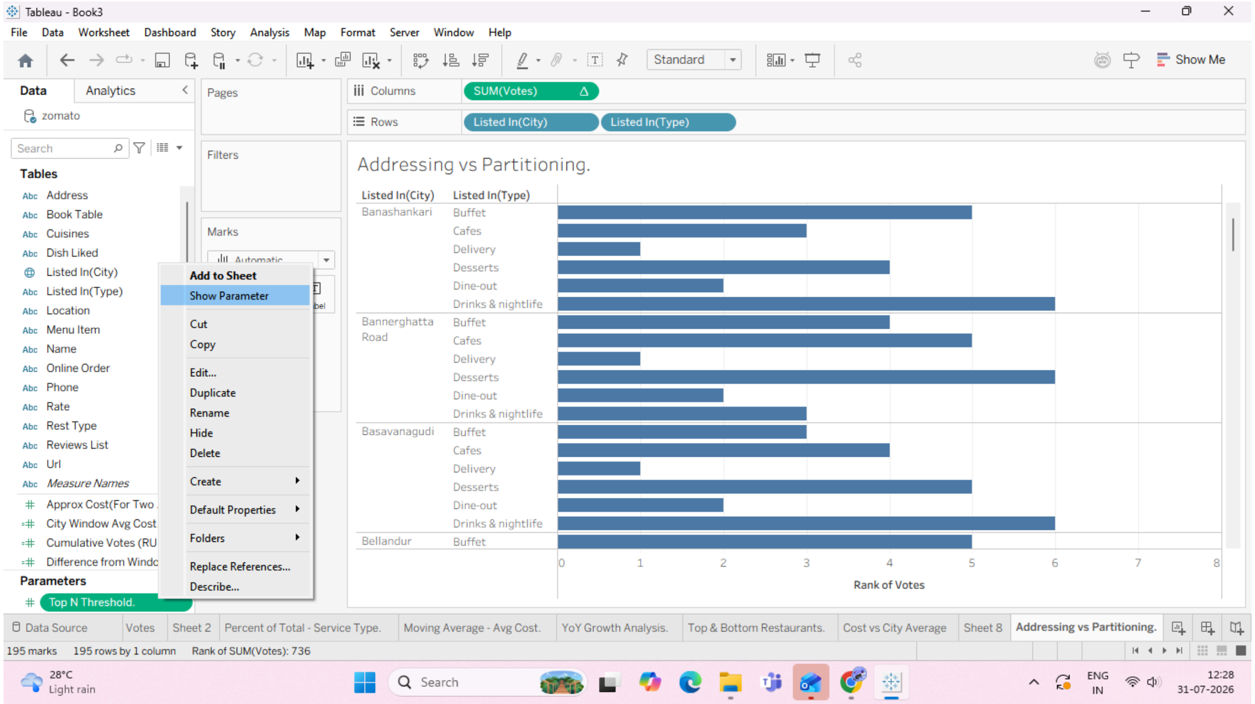


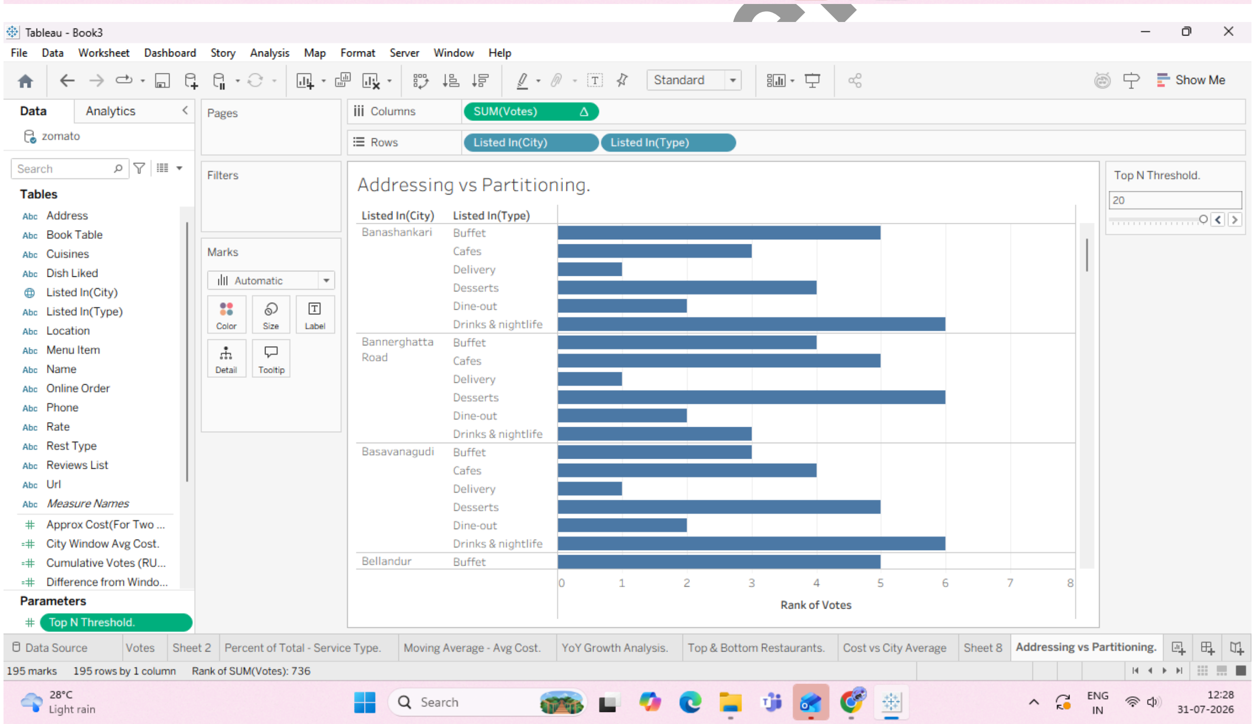


7. Right-click Top N Threshold in the Data Pane and select Show Parameter Control.

Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

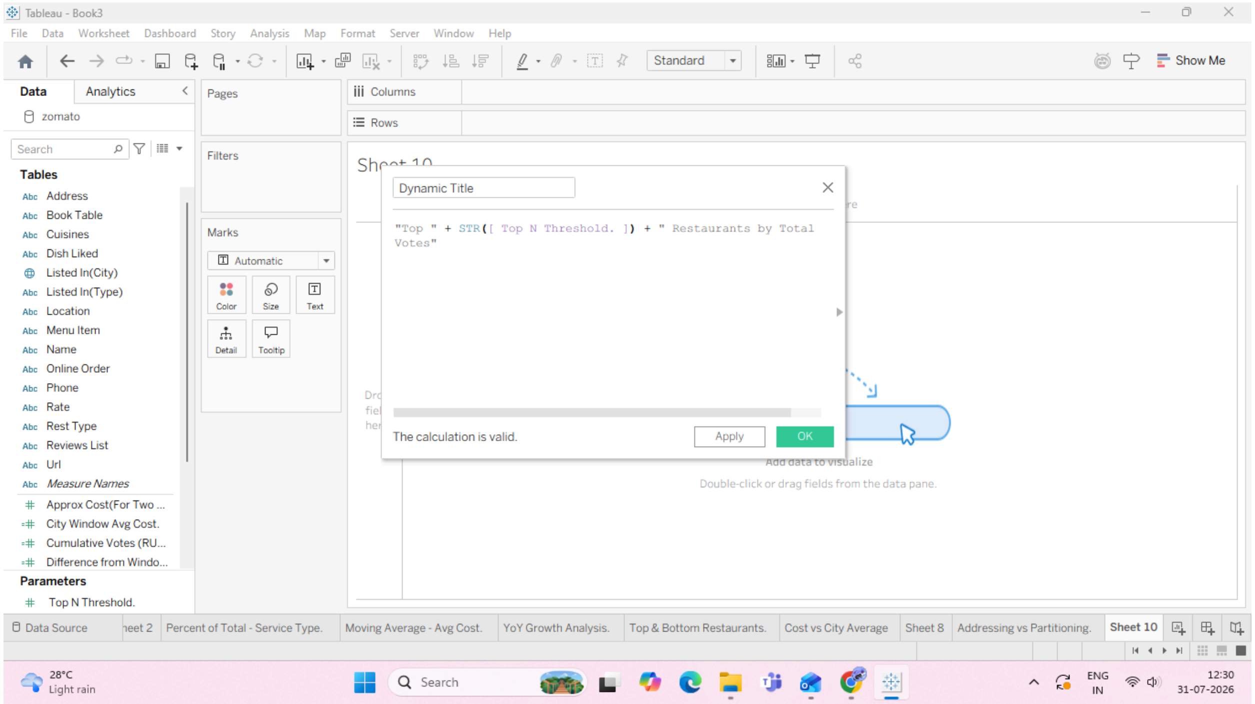




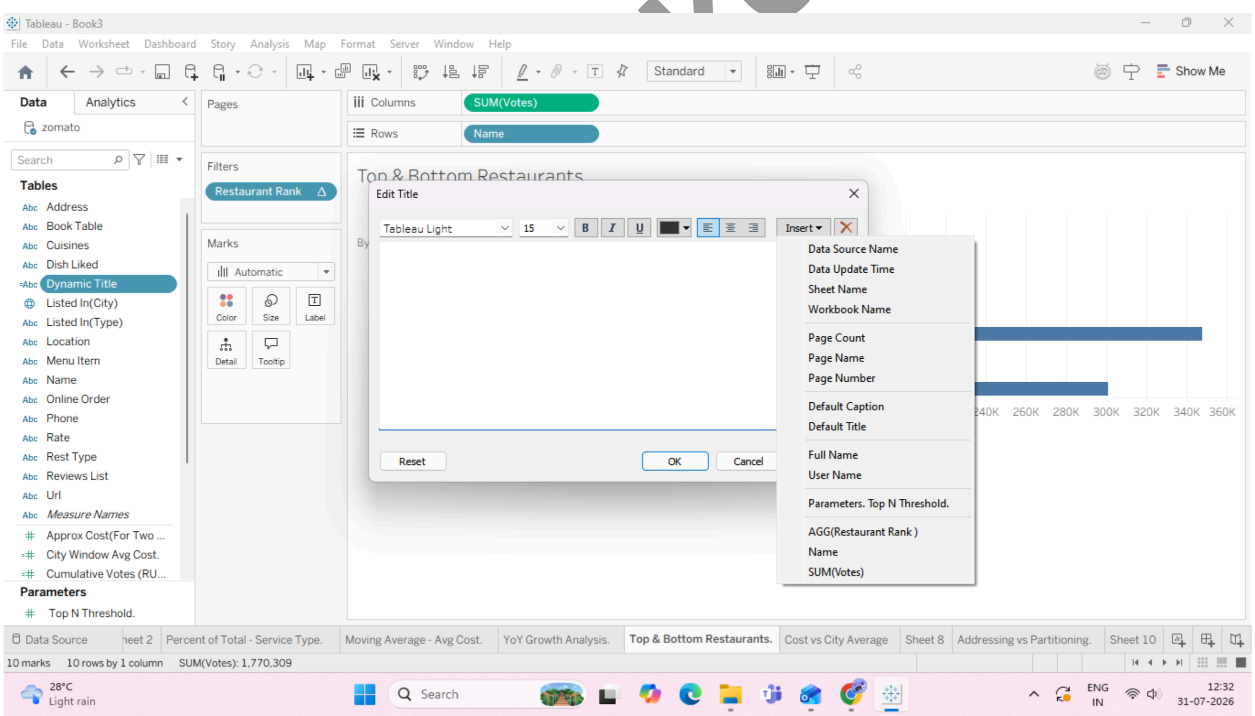
Task 2: Create a Dynamic Header Calculation

Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

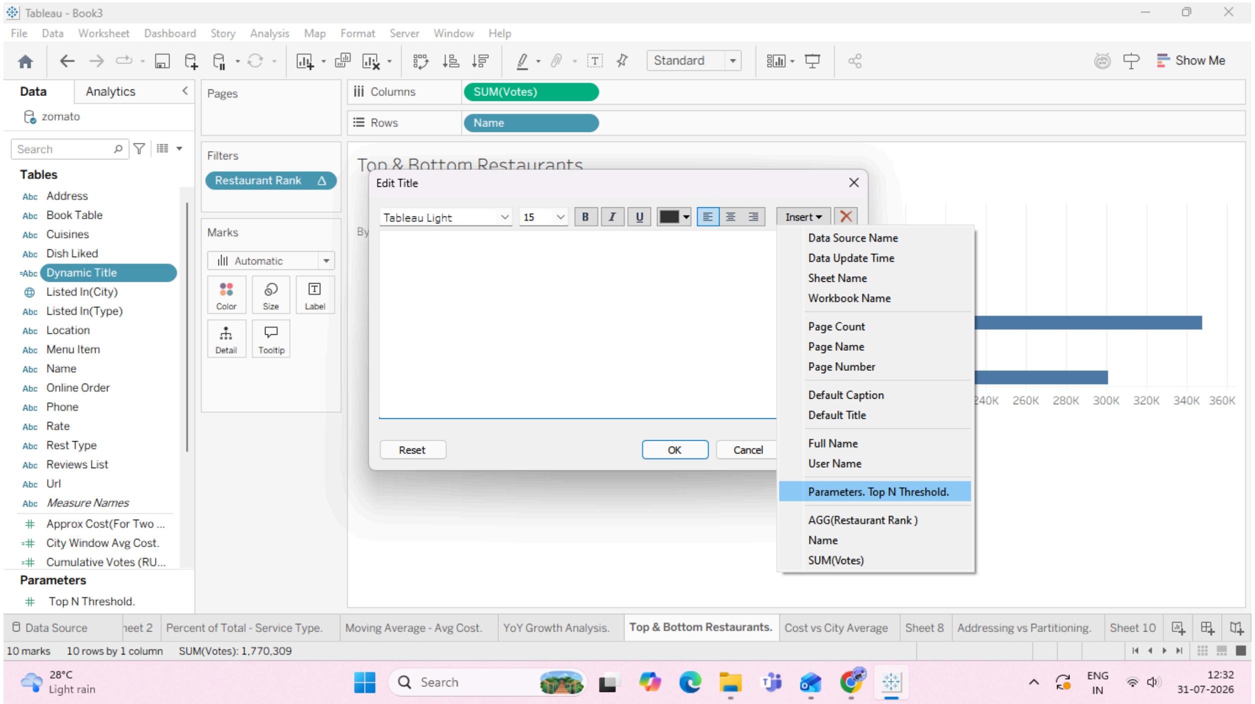


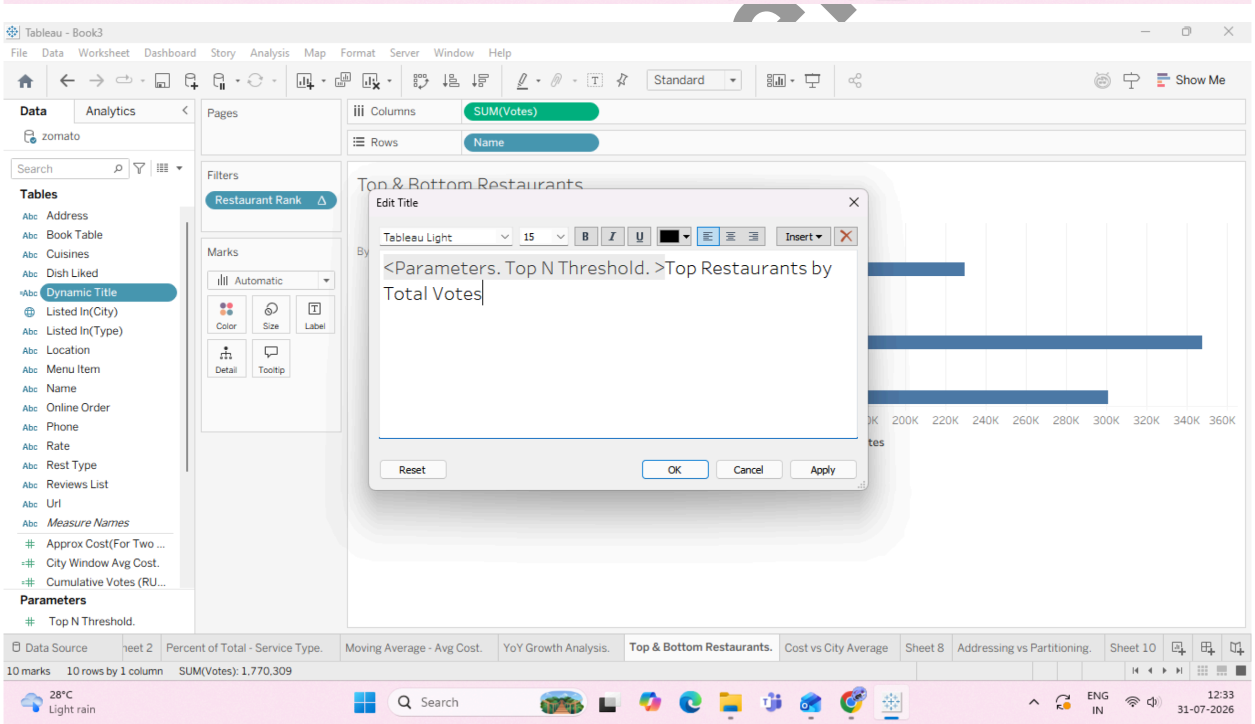
Task 3: Embed Calculated Fields and Parameters into the Sheet Title



Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

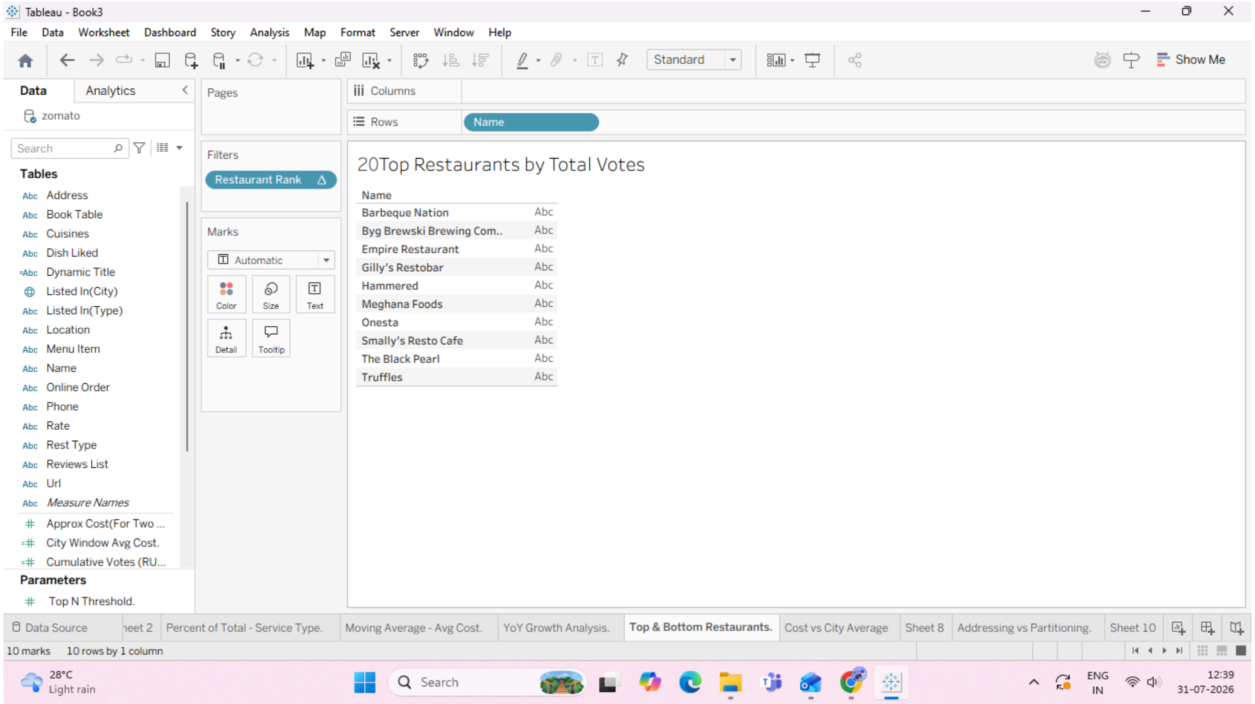




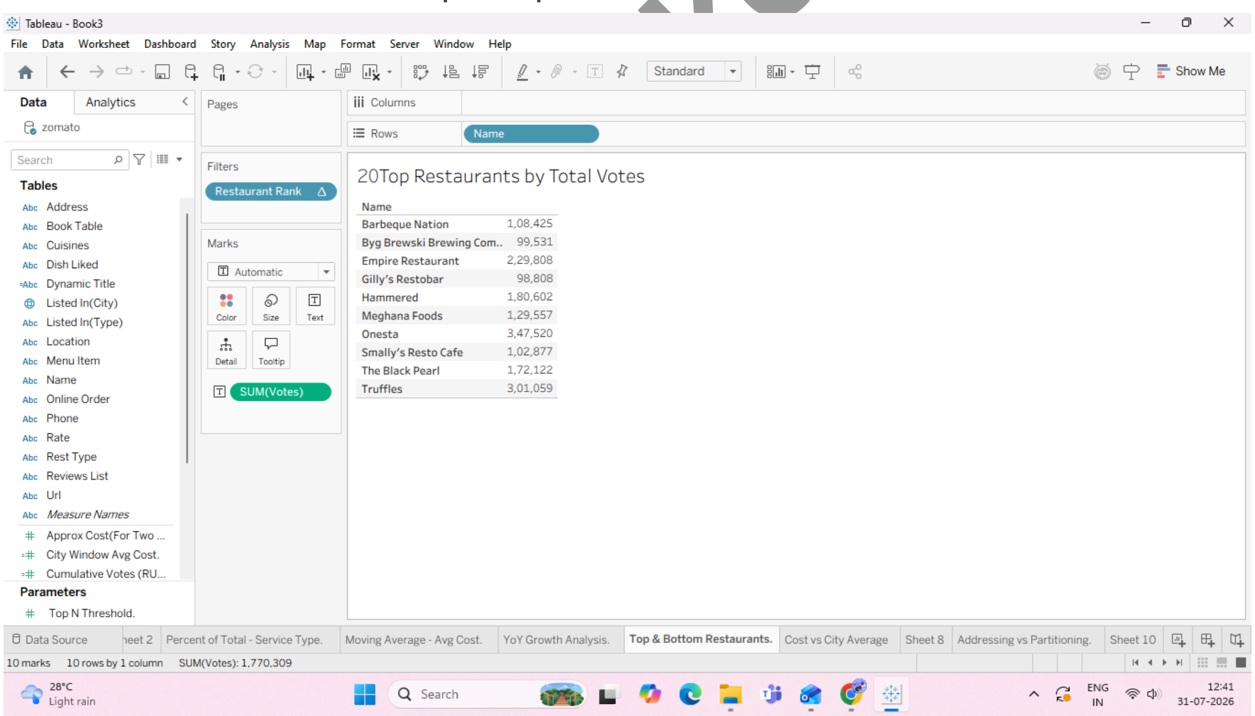
Task 4: Add Dynamic Mark Labels (Highlighting Extremes)

Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE



2. Click the Label card to open options:



Purvi.R.Karkera
T086